1. Heredity and Evolution

- Heredity and Hereditary Changes
- Evolution
- Darwin's Theory of Natural Selection
- Speciation
- Transcription, Translation & Translocation
- Evidences of Evolution
- Lamarckism
- Human Evolution

Can you recall?

1. Which component of the cellular nucleus of living organisms carries hereditary characters?
2. What do we call to the process of transfer of physical and mental characters from parents to the progeny?
3. Which are the components the DNA molecule?

Heredity and hereditary changes

You know that heredity is the transfer of biological characters from one generation to another via genes. Gregor Johann Mendel is pioneer of the modern genetics. It took a long time to understand the conclusions of his research about heredity. In 1901, the reasons behind the sudden changes were understood due to the mutational theory of Hugo de Vries. Meanwhile in 1902, Walter Sutton observed the paired chromosomes in the cells of grasshopper; until then it was not known to anyone. Research started in the direction of finding the nature of genetic material when it was proved that genes are carried via chromosomes. Through which 1944, trio of scientists Oswald Avery, Mclyn McCarty and Colin McLeod proved that except viruses, all living organisms have DNA as genetic material.

In 1961, the French geneticists Francois Jacob and Jack Monod proposed a model for process of protein synthesis with the help of DNA in bacterial cells. It helped to uncover the genetic codes hidden in DNA. Thereby, the technique of recombinant DNA technology emerged which has vast scope in the field of genetic engineering.

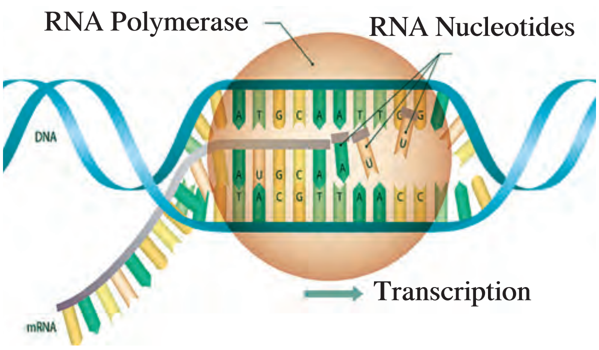
The science of heredity is useful for diagnosis, treatment and prevention of hereditary disorders, production of hybrid varieties of animals and plants and in industrial processes in which microbes are used.

Transcription, Translation and Translocation

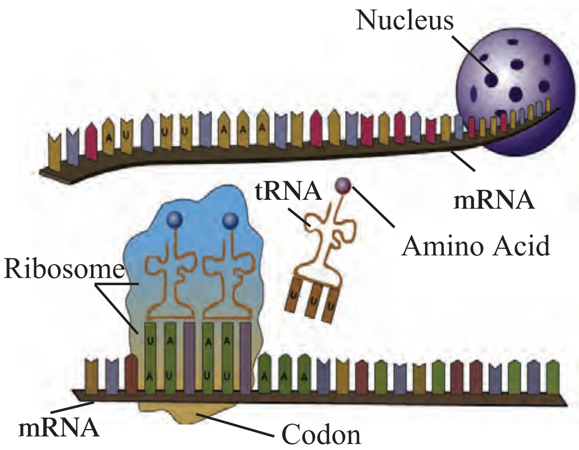
Can you tell?

1. Sketch and explain the structure of DNA and various types of RNA.
2. Explain the meaning of genetic disorders and give names of some disorders.

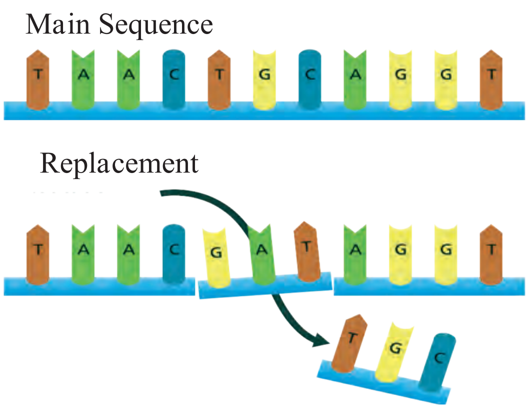
With the help of RNA, the genes present in the form of DNA participate in the functioning of cell and thereby control the structure and functioning of the body. Information about protein synthesis is stored in the DNA and synthesis of appropriate proteins as per requirement is necessary for body. These proteins are synthesized by DNA through the RNA. This is called as 'Central Dogma'. mRNA is produced as per the sequence of nucleotides on DNA. Only one of the two strands of DNA is used in this process. The sequence of nucleotides in mRNA being produced is always complementary to the DNA strand used for synthesis. Besides, there is uracil in RNA instead of thymine of DNA. This process of RNA synthesis is called as 'transcription'.



1.1 Transcription



1.2 Translation & Translocation



1.3 Mutation

\` The mRNA formed in nucleus comes in cytoplasm. It brings in the coded message from DNA. The message contains the codes for amino acids. The code for each amino acid consists of three nucleotides. It is called as 'triplet codon'.

Dr Har Govind Khorana, a scientist of Indian origin has made an important contribution in discovery of triplet codons for 20 amino acids. For this work, he has been awarded with the Nobel Prize in 1968, along with two other scientists.

Each mRNA is made up of thousands of triplet codons. As per the message on mRNA, amino acids are supplied by the tRNA. For this purpose, tRNA has 'anticodon' having complementary sequence to the codon on mRNA. This is called as 'translation'. The amino acids brought in by tRNA are bonded together by peptide bonds with the help of rRNA. During this process, the ribosome keeps on moving from one end of mRNA to other end by the distance of one triplet codon. This is called as 'translocation'. Such many chains come together to form complex proteins. These proteins control various functions in the body of living organisms and their appearance too.

Living organisms can produce new individuals like themselves due to genes only and some of those genes are transmitted to the next generation without any changes. Due to this, some of the characters of parents are transmitted to their offsprings. However, sometimes sudden changes occur in those genes. Sometimes, any nucleotide of the gene changes its position that causes a minor change which is nothing but the 'mutation'.

Some mutations may be minor but some may be considerable. Ex. Mutation may cause the genetic disorders like sickle cell anaemia.

Can you recall?

1. What is the function of the appendix of our digestive system?
2. Are our wisdom teeth really useful for chewing the food?
3. Why did the huge animals like dinosaur become extinct?
4. Why are many species of animals and birds getting extinct?

Evolution

Evolution is the gradual change occurring in living organisms over a long duration. This is a very slow-going process through which development of organisms is achieved. All the stages in changes occurred in various components ranging from stars and planets in space to the biosphere present on the Earth should be included in the study of evolution. Formation of new species due to changes in specific characters of several generations of living organisms as a response to natural selection, is called as evolution.

3.5 billion years ago, life had been non existent on the Earth. At the beginning, there may have been only simple elements in the ocean on the Earth and simple type of organic and inorganic compounds may have been formed from those. Complex compounds like proteins and nucleic acids may have formed over the long period from those simple compounds. First primitive type of cells may have been formed from the mixture of different types of organic and inorganic compounds. Number of those cells may have increased at the cost of surrounding chemicals. There may had been some differences among those cells and according to the principle of natural selection, some may have shown good growth and some may have perished which could not adjust with the surrounding.

At present, crores of species of plants and animals with huge diversity regarding shape and complexity are present on the Earth. Animal diversity ranges from the unicellular *Amoeba* and *Paramecium* to man and giant whale. The plant diversity consists of various species ranging from unicellular *Chlorella* to the huge banyan tree. The life exists on Earth everywhere from equator to both the poles. Organisms are present at all the places like air, water, land, rock, etc. Humans have shown curiosity about origin of life and reasons for such a great diversity in life present on the Earth since ancient period. Different theories about origin and evolution of life have been proposed till today of which theory of 'Gradual development of living organisms is accepted.

Internet is my friend

Collect the information from internet about Big-Bang theory related with the formation of stars and planets and present it in your class.

A peek into History

Many philosophers and religious scholars have written their views about formation of life. There seems to be a thorough discussion over the formation of Universe, in various cultures like Indian, Chinese, Roman, Greek, etc. Various cultures have noted different type of information about planets, stars, the 'panchmahabhuta', living organisms, etc. in the form of poetry, stories and religious / sacred books.

Theory of Evolution:

According to this theory, first living material (protoplasm) has been formed in ocean. In due course of time, unicellular organism was formed. Gradually, changes occurred in the unicellular organisms from which larger and more complex organisms were formed. All those changes were slow and gradual. Duration of all these changes is at most 300 crore years. Changes and development in living organisms had been all round and multi-dimensional and this led to evolution of different types of organisms. Hence, this overall process is called as evolution which is organizational. Progressive development of plants and animals from the ancestors having different structural and functional organization is called evolution.

Evidences of evolution

Collective thinking upon all above mentioned theories implies that evolution is everlasting process of changes. However, it needs proof to prove it. Following are various proofs available in support of the theories mentioned above.

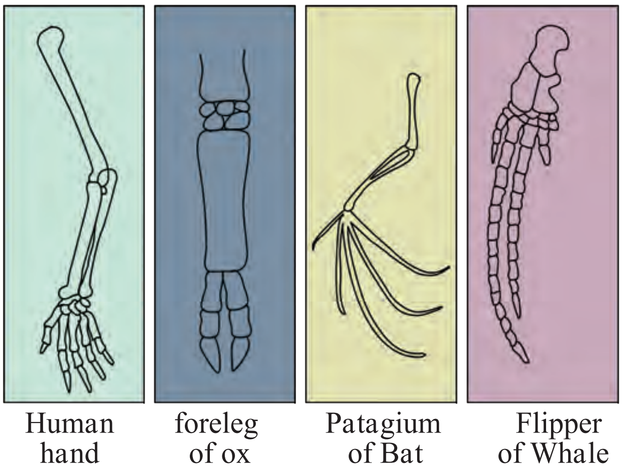
1. Morphological Evidences

Try this

Observe the following images and note the similarities between given animal images and plant images.

Various similarities like structure of mouth, position of eyes, structure of nostrils and ear pinnae and thickly distributed hairs on body are seen in animals whereas similarities in characters like leaf shape, leaf venation, leaf petiole, etc. occur in case of plants. This indicates that there are some similarities in those groups and hence it proves that their origin must be same and must have common ancestors

1.4 Morphological evidences



1.5 Structure of bones

2. Anatomical Evidences

If you carefully observe the pictures, there doesn't seem any superficial similarity between human hand, foreleg of ox, flipper of whale and patagium of bat. Similarly, use of each of those structures is different in respective animals. However, there is similarity in structure of bones and joints in organs of each of those animals. This similarity indicates that those animals may have common ancestor.

Can you tell?

1. Which are the different organs in body of organisms?
2. Is each of the organs useful to organism?

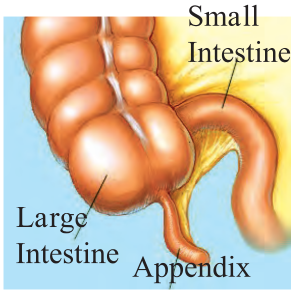
Use of ICT :

Collect the information of geological dating and Present it classroom.

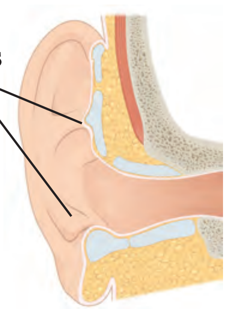
3. Vestigial Organs

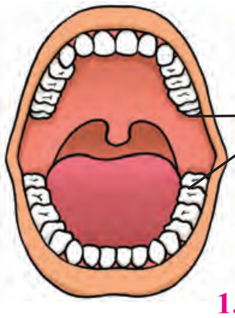
Degenerated or underdeveloped useless organs of organisms are called as vestigial organs. In living organisms, sudden development of new tissues or organs for living in changing environment is not possible. Instead, existing organs undergo gradual changes. Mostly, a specific structure in the body is useful under certain situation. However, same structure under different situation may become useless or even harmful. Such structure begins to degenerate under such situation as per the principle of natural selection. It takes thousands of years for a structure to disappear. Such organs are seen in different phases of disappearance in different animals. Such organ, though non-functional in certain organisms, it may be functional in other organisms i.e. it is not vestigial in other organisms.

Appendix, which is useless to human, is useful and fully functional organ in ruminants. Similarly, muscles of ear pinna, which are useless to human, are useful in monkeys for movement of ear pinna. Various vestigial organs like tail-bone (coccyx), wisdom teeth, and body hairs are present in body of human being.



Ear muscles





Tail bone (Coccyx)

1.6 Vestigial organs

Observe and discuss. Observe the following pictures.

1.7 Some fossils

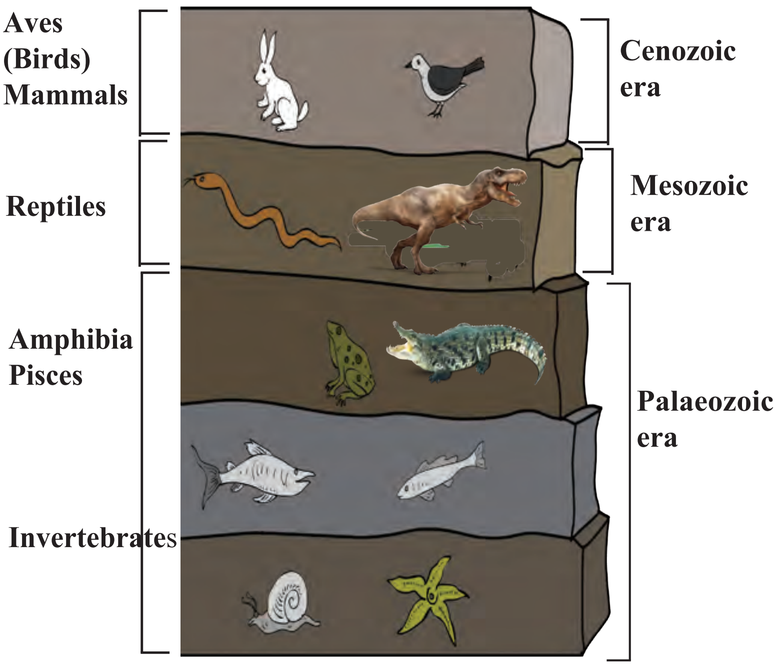
Use of ICT

Find how the vestigial organs in certain animals are functional in others. Present the information in your class and send it to others.

4. Palaeontological Evidences

A question may arise in your mind that which organisms existed millions of years ago? How can we tell this? Now this secret has been hidden in the Earth. Large number of organisms get buried due to disasters like flood, earthquake, volcano, etc. Remnants and impressions of such organisms remain preserved underground. These are called as fossils. Study of fossils is an important aspect of study of evolution.

Carbon consumption of animals and plants stops after death and since then, only the decaying process of C-14 occurs continuously. In case of dead bodies of plants and animals, instead of remaining constant, the ratio between C-14 and C-12 changes continuously as C-12 is non-radioactive. The time passed since the death of a plant or animal can be calculated by measuring the radioactivity of C-14 and ratio of C-14 to C-12 present in their body. This is 'carbon dating' method. It is used in palaeontology and anthropology for determining the age of human fossils and manuscripts. Once the age of fossil been determined by such technique, it becomes easy to deduce the information about other erstwhile organisms. It seems that vertebrates have been slowly originated from invertebrates.



1.8 Sedimentary rocks and fossils

Introduction to Scientists

Carbon dating method is based upon the radioactive decay of naturally occurring C-14 and it is developed by Willard Libby.

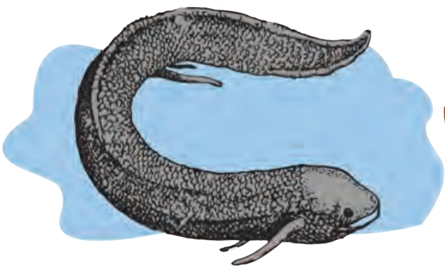
He has been awarded with Nobel Prize (1960) for this invention. The age of the materials determined by this method are published in the journal 'Radio Carbon'

5. Connecting Links

Observe and discuss.

Observe the following pictures and discuss the characters observed.

Duckbill Platypus



Lungfish

Peripatus

1.9 Some animals with special characteristics

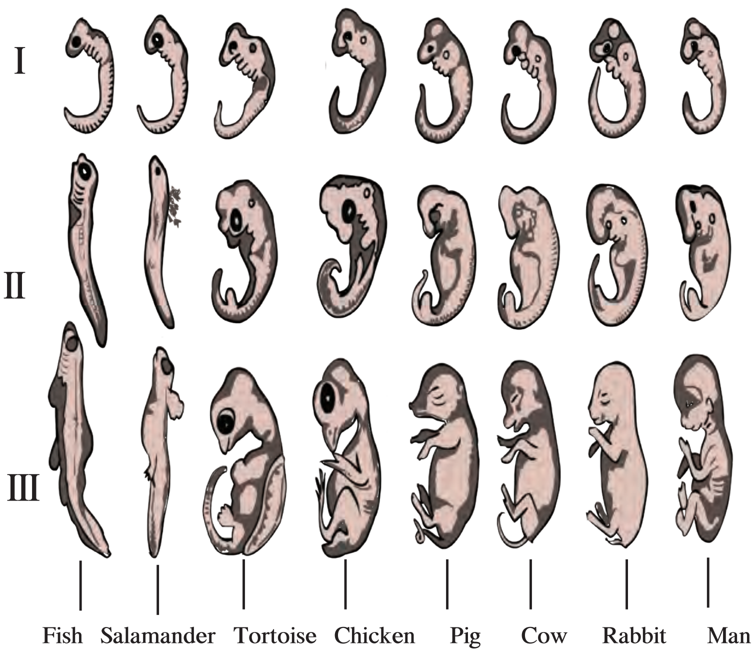
Some plants and animals show some morphological characters by which they can be related to two different groups; hence they are called as 'connecting links'. Ex. In *Peripatus*, characters like segmented body, thin cuticle, and parapodia-like organs are present. Similarly, these animals show tracheal respiration and open circulatory system similar to arthropods. This indicates that *Peripatus* is connecting link between annelida and arthropoda. Similarly, duck billed platypus lays eggs like reptiles but shows relationship with mammals too due to presence of mammary glands and hairs. Lung fish performs respiration with lungs irrespective of being fish. These examples indicate that mammals are evolved from reptiles and amphibians from fishes.

Observe and discuss.

Carefully observe the stages of embryonic development of some animals shown in fig. 1.10

6. Embryological Evidences:

Comparative study of embryonic developmental stages of various vertebrates given in the picture shows that all embryos show extreme similarities during initial stages and those similarities decrease gradually. Similarities in initial stages indicate the common origin of all these animals.



1.10 Embryos during different stages

Darwin's theory of natural selection

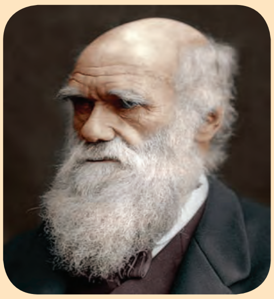
Charles Darwin had collected innumerable specimens of plants and animals and depending upon the observations of those specimens; he published the theory of natural selection which preaches the survival of fittest. For this purpose, Darwin had published a book titled **'Origin of Species'**. While explaining the concept, Darwin says that all the organisms reproduce prolifically. All the organisms compete with each other in a life--threatening manner. In this competition, only those organisms sustain which show the modifications essential for winning the competition. However, besides this, natural selection also plays important role because nature selects only those organisms which are fit to live and the rest perish. Sustaining and selected organisms can perform reproduction and thereby give rise to the new species with their own specific characters. Darwin's theory of natural selection was widely accepted for long duration. However, some objections were raised against the theory. Some of the main objections are-

1. Natural selection is not the only factor responsible for evolution.
2. Darwin did not mention any explanation about useful and useless modifications.
3. There is no explanation about slow changes and abrupt changes.

Irrespective of all these objections, Darwin's work on evolution has been a milestone.

Introduction to Scientists

Charles Robert Darwin (1809-1882)

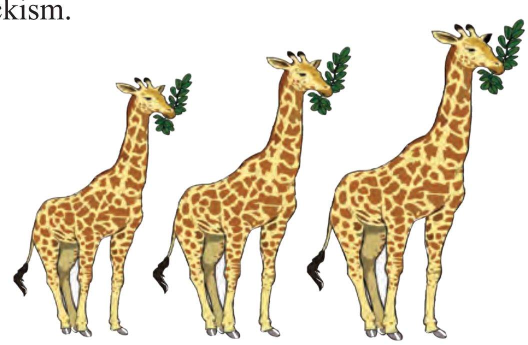


This English biologist proposed the theory of evolution. He showed that all the species of living organisms have been gradually evolved over the period of thousands of years from common ancestor. He proposed that principle of natural selection is responsible for this evolution.

Lamarckism

Jean-Baptiste Lamarck proposed that morphological changes occurring in living organisms are responsible for evolution and the reason behind those morphological changes is activities or laziness of that organism. He called this concept as principle of 'use or disuse of organs'.

Further, he said that the neck of giraffe has become too long due to browsing on leaves of tall plants by extending their neck for several generations; similarly, shoulders of the ironsmith have become very strong due to frequent hammering movements. Wings of birds like ostrich and emu have become weak due to no use. Legs of the birds like swan and duck have become useful for swimming due to living in water and snakes have lost their legs by modifications in their body for burrowing habit. All these examples are types of 'acquired characters' and are transferred from one to another generation. This is called as theory of inheritance of acquired characters or Lamarckism.



1.11 Giraffe

Development of organs due to specific activities or their degeneration due to no use at all was widely accepted but transfer of those characters from generation to generation was rejected. Because it had been verified many times that modifications brought in us are not transferred to next generation and thereby Lamarck's theory was disproved.

The living organism can transfer the characters which it has acquired, to the next generation. This is called ancestry of acquired characters.

Introduction to Scientists

Jean-Baptiste Lamarck (1744-1829)

Lamarck proposed that the activities of the organisms are responsible for their evolution. This French naturalist proposed that each animal or plant undergo some changes in its life span and those changes are transferred to the next generation and such changes occur in next subsequent generations too.

Internet is my friend

Collect the pictures and information of various species of monkeys from internet.

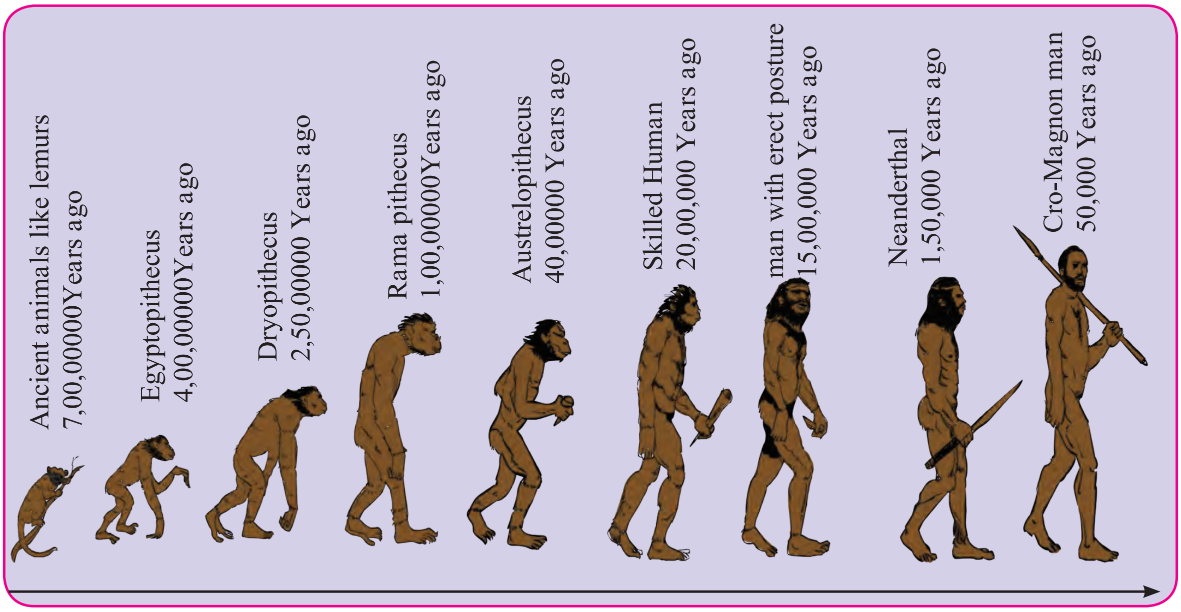
Speciation

Formation of new species of plants and animals is the effect of evolution. Species is the group of organisms that can produce fertile individuals through natural reproduction. Each species grows in specific geographical conditions. Their food, habitat, reproductive ability and period is different. However, genetic variation is responsible for formation of new species from earlier one. Besides, geographical and reproductive changes are also responsible. Similarly, geographical or reproductive isolation also leads to speciation

Human Evolution

The biodiversity that is known today has been said to be formed from very simple unicellular organism due to evolution. In this evolution, origin of human evolution can be shown as per the picture given below. Last dinosaurs disappeared approximately seven crore years ago. At that time, monkey-like animals are said to be evolved from some ancestors who were more or less similar to the modern lemurs. Tail of these monkey-like animals of Africa is said to be disappeared about 4 crore years ago. They developed due to enlargement in brain their hands were also improved and thus ape-like animals were evolved. Meanwhile, these ape-like animals reached the South and North-East Asia and finally evolved into gibbon and orang-utan.

Remaining ape-like animals stayed in Africa and from them, gorilla and chimpanzee evolved about 2.5 crore years ago. Evolution of some of the 2 crore year old species of apes seems to be occurred in different way. They had to use their hands more for eating food and other work.

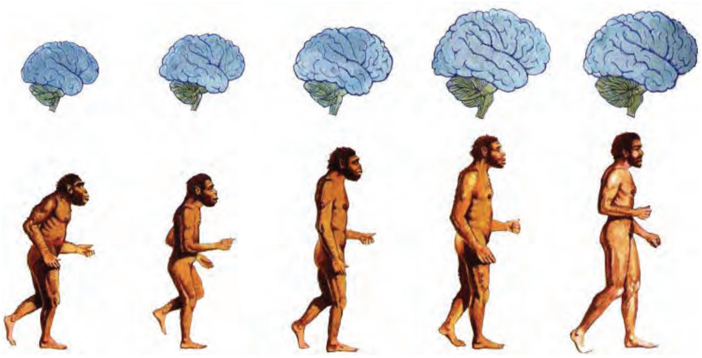


1.12 Journey of human

Those apes started to live on land as the forests started to decline due to dry environment. Their pelvic girdle developed in such a way that they started to stand in erect posture in grasslands and thereby their hands became available for use, anytime. These first human--like animals with erect posture which were using their hands have evolved about 2 crore years ago.

First record of human-like animal is with us in the form of 'Ramapithecus' ape from North India and East Africa. Afterwards, this ape grown up in size and became more intelligent and thus the ape of South Africa evolved about 40 lakh years ago.

The morphology of these human-like animals started to appear like to be the member of the genus *Homo*, about 20 lakh years ago and thus skilled human developed. About 15 lakh years ago, human walking with erect posture was evolved. It may have existed in China and Indonesia of Asian continent.



1.13 Development of human brain

Evolution of upright man continued in the direction of developing its brain for the period of about 1 lakh years and meanwhile it discovered the fire. Brain of 50 thousand year old man had been sufficiently evolved to the extent that it could be considered as member of the class- wise-man (*Homo sapiens*).

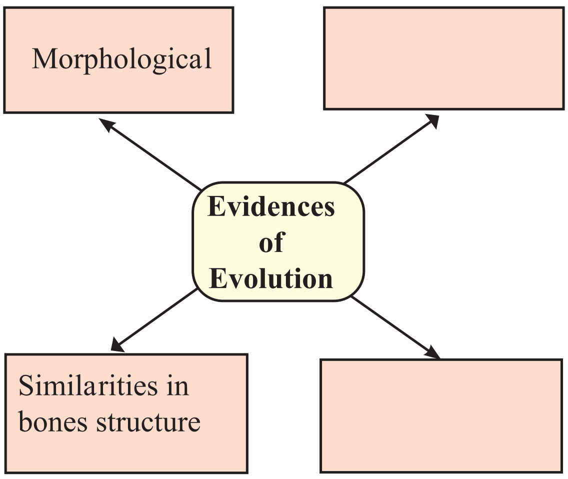
Neanderthal man can be considered as the first example of wise-man. The Cro-Magnon man evolved about 50 thousand years ago and afterwards, this evolution had been faster than the earlier.

1.14 Neanderthal man

About 10 thousand years ago, wise-man started to practice the agriculture. It started to rear the cattle-herds and established the cities. Cultural development took place. Art of writing was invented about 5000 years ago and thus the history had been started. Modern sciences emerged about 400 years ago and industrial society was established about 200 years ago and now we have reached at this stage, and still we are searching the details of roots of human ancestry.

Exercise

1. Complete the following diagram.



2. Read the following statements and justify the same in your own words with the help of suitable examples.

a. Human evolution began approximately 7 crore years ago.

b. Geographical and reproductive isolation of organisms gradually leads to speciation.

c. Study of fossils is an important aspect of study of evolution.

d. There is evidence of fatal science among chordates.

3. Complete the statements by choosing correct options from bracket.

(Gene, Mutation, Translocation, Transcription, Gradual development, Appendix)

a. The causality behind the sudden changes was understood due to -- -- principle of Hugo de Vries.

b. The proof for the fact that protein synthesis occurs through -- --- was given by George Beadle and Edward Tatum.

c. Transfer of information from molecule of DNA to mRNA is called as -- -- -- process.

d. Evolution means -- -- -- --.

e. Vestigial organ -- -- -- present in human body is proof of evolution.

4. Write short notes based upon the information known to you.

a. Lamarckism

b. Darwin's theory of natural selection.

c. Embryology.

d. Evolution.

e. Connecting link.

5. Define heredity. Explain the mechanism of hereditary changes.

6. Define vestigial organs. Write names of some vestigial organs in human body and write the names of those animals in whom same organs are functional.

7. Answer the following questions.

a. How are the hereditary changes responsible for evolution?

b. Explain the process of formation of complex proteins.

c. Explain the theory of evolution and mention the proof supporting it.

d. Explain with suitable examples importance of anatomical evidences in evolution.

e. Define fossil. Explain importance of fossils as proof of evolution.

f. Write evolutionary history of modern man.

Project :

1. Make a presentation on human evolution using various computer softwares and arrange a group discussion over it in the class room.
2. Read the book – 'Pruthvivar Manus Uparach' written by Late Dr. Sureshchandra Nadkarni and note your opinion on evolution.

KLG6EJ

2. Life Processes in Living Organisms Part -1

- Living Organisms & Life Processes
- Living Organisms & Energy Production
- Some Nutrients & Energy Efficiency
- Cell Division- A Life Process

Can you recall?

1. How are the food stuffs and their nutrient contents useful for body?
2. What is the importance of balanced diet for body?
3. Which different functions are performed by muscles in body?
4. What is the importance of digestive juices in digestive system?
5. Which system is in action for removal of waste materials produced in human body?
6. What is the role of circulatory system in energy production?
7. How are the various processes occurring in human body controlled? In how many ways ?

Living Organisms and Life Processes

Various organ-systems are continuously performing their functions in human body. Along with the various systems like digestive, respiratory, circulatory, excretory and control systems, different external and internal organs are performing their functions independently but through a complete co-ordination. This overall system is in action in more or less same way in all the organisms. Those are in need of continuous source of energy for this purpose. Carbohydrates, fats and lipids are the main sources of this energy and it is harvested by the mitochondria present in each cell. It is not like that only foodstuff is sufficient for energy production but oxygen is also necessary. All these i.e. food stuffs and oxygen are transported up to the cell via circulatory system. Besides, it is coordinated by the control system of the body. i.e. each life process contributes in its own way in the process of energy production. Functioning of all these life processes also requires the energy.

Human and other animals consume the fruits and vegetables. Plants are autotrophs. They prepare their own food. They utilize some of the food for themselves whereas remaining is stored in various parts like fruits, leaves, stem, roots, etc. We consume all these various plant materials and obtain different nutrients like carbohydrates, fats, proteins, vitamins, minerals, etc. Which food materials do we consume to obtain these nutrients?

We obtain the carbohydrates from milk, fruits, jaggary, cane sugar, vegetables, potatoes, sweet potatoes, sweetmeats and cereals like wheat, maize, ragi, jowar, millet, rice, etc. We get 4Kcal energy per gram of carbohydrates. Let us study the way by which this energy is obtained.

Many players are seen consuming some food stuffs during breaks of the game.

Use your brain power

Why may be the players consuming these food stuffs?

Can you recall?

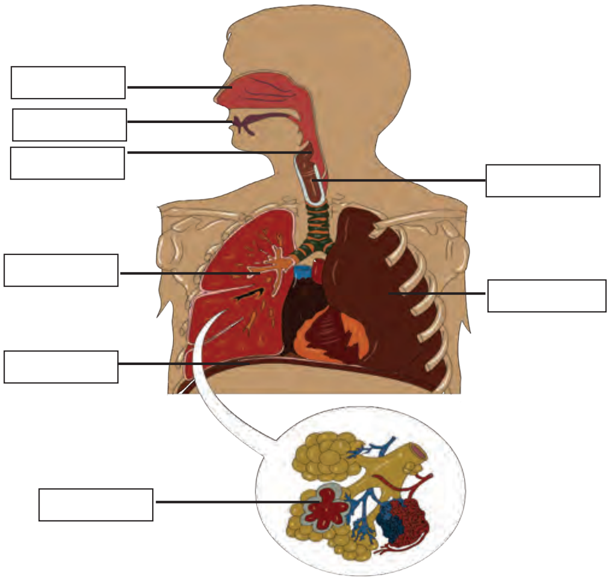
1. What is respiration? How does it occur?

Living organism and Energy production

Observe

Observe and Label the diagram given beside.

In living organisms, respiration occurs at two levels as body and cellular level. Oxygen and carbon dioxide are exchanged between body and surrounding in case of respiration occurring at body level. In case of respiration at cellular level, foodstuffs are oxidized either with or without help of oxygen.



2.1 Human respiratory system

Can you tell?

1. How many atoms of C, H and O are respectively present in a molecule of glucose?
2. Which types of chemical bonds are present between all these atoms?
3. In terms of Chemistry what happens actually when a molecule is oxidized?

Carbohydrates of the food that we consume everyday are mainly utilized for production of energy required for daily need. This energy is obtained in the form of ATP. For this purpose, glucose, a type of carbohydrates is oxidized step by step in the cells. This is called as cellular respiration. Cellular respiration occurs among the living organisms by two methods. Those two methods are aerobic respiration (oxygen is involved) and anaerobic respiration (oxygen is not involved). In aerobic respiration, glucose is oxidized in three steps.

1. Glycolysis

Process of glycolysis occurs in cytoplasm. A molecule of glucose is oxidized step by step in this process and two molecules of each i.e. pyruvic acid, ATP, $NADH_2$ and water are formed.

Molecules of pyruvic acid formed in this process are converted into molecules of Acetyl-Coenzyme-A. Two molecules of $NADH_2$ and two molecules of CO_2 are released during this process.

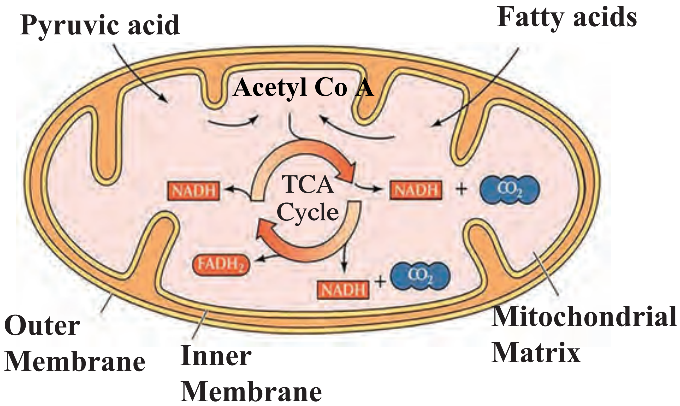
2. Tricarboxylic acid cycle (Krebs cycle)

Both molecules of acetyl-CoA enter the mitochondria. Cyclic chain of reactions called as tricarboxylic acid cycle is operated on it in the mitochondria. Acetyl part of acetyl-CoA is completely oxidized through this cyclical process and molecules CO_2, H_2O, $NADH_2$, $FADH_2$ are derived.

3. Electron transfer chain reaction

Molecules of $NADH_2$ and $FADH_2$ formed during all above processes participate in electron transfer chain reaction. Due to this, 3 molecules of ATP are obtained from each $NADH_2$ molecule and 2 molecules of ATP from each $FADH_2$ molecule. Besides ATP, water molecules are also formed in this reaction. Electron transfer chain reaction is operated in mitochondria only.

Thus, a molecule of glucose is completely oxidized in aerobic respiration and molecules of CO_2 and H_2O are produced along with energy.



2.2 Mitochondria and Tri-carboxylic acid cycle

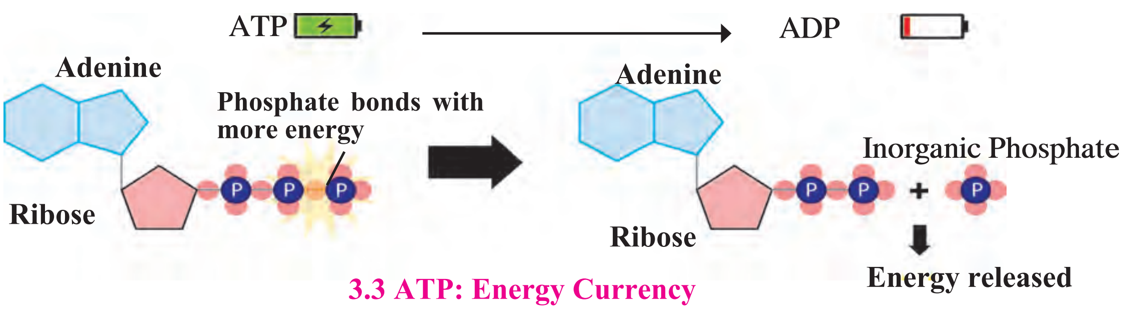
Always Remember.

NAD - Nicotinamide Adenine dinucleotide

FAD - Flavin adenine dinucleotide

Both coenzymes are formed in the cells and used in cellular respiration.

ATP: Adenosine triphosphate is energy-rich molecule and energy is stored in the bonds by which phosphate groups are attached to each other. These molecules are stored in the cells as per need. Chemically, ATP is triphosphate molecule formed from adenosine ribonucleoside. It contains a nitrogenous compound-adenine, pentose sugar- ribose and three phosphate groups. As per the need, energy is derived by breaking the phosphate bond of ATP; hence ATP is called as 'energy currency' of the cell.



3.3 ATP: Energy Currency

If there is insufficient amount of carbohydrates in body due to exceptional conditions like fasting and hunger, then lipids and proteins are used for energy production. In case of lipids, they are converted into fatty acids whereas proteins into amino acids. Fatty acids and amino acids are converted into acetyl-CoA and energy is obtained through complete oxidation of acetyl-CoA by the process of Krebs cycle in mitochondria.

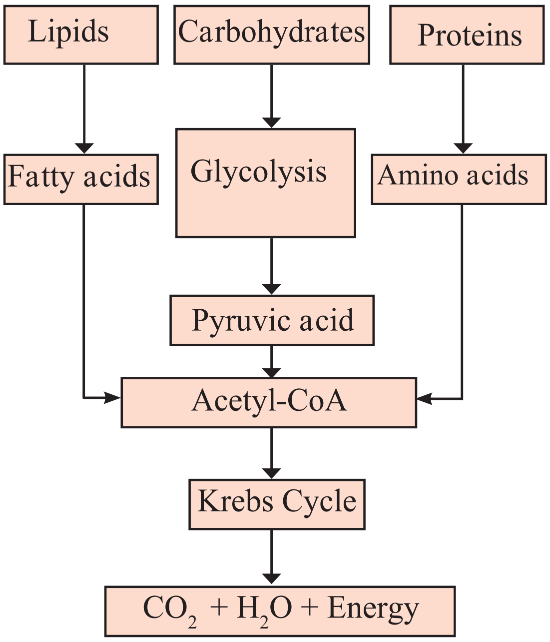
Introduction To Scientists

Sir Hans Krebs
(1900-1981)

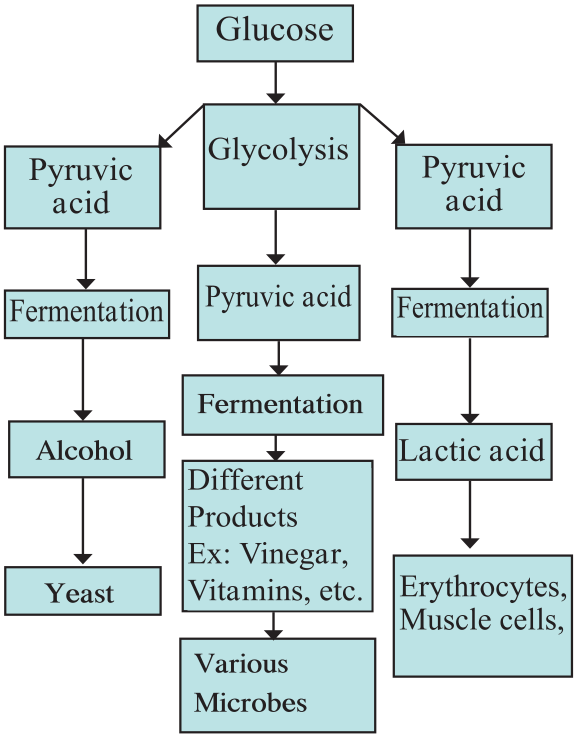
Process of glycolysis was discovered by three scientists Gustav Embden, Otto Meyerhof, and Jacob Parnas along with their colleagues. For this purpose, they performed experiments on muscles. Hence, glycolysis is also called as Embden-Meyerhof-Parnas pathway (EMP pathway).

The cyclical reactions of tricarboxylic acid cycle were discovered by Sir Hans Krebs. Hence, this cyclical process is also called as Krebs cycle. He has been awarded the Nobel Prize in 1953 for this discovery.

Process of energy production through aerobic respiration of carbohydrates, proteins and fats.



Anaerobic respiration in living organisms/ cells



Energy Production in Microorganisms through Anaerobic Respiration

Some organisms cannot live in presence of oxygen. Ex. Many bacteria. Such living organisms have to perform anaerobic respiration for energy production. Glycolysis and fermentation are two steps of anaerobic respiration. Glucose is incompletely oxidized and less amount of energy is obtained in this type of respiration. Pyruvic acid produced through glycolysis is converted into other organic acids or alcohol with the help of some enzymes in this process. This is called as fermentation. Some higher plants, animals and aerobic microorganisms also perform anaerobic respiration instead of aerobic respiration if there is depletion in oxygen level in the surrounding.

Ex. Seeds perform anaerobic respiration if the soil is submerged under water during germination. Similarly, our muscle cells also perform anaerobic respiration while performing the exercise. Due to this, less amount of energy is produced in our body and lactic acid accumulates due to which we feel tired.

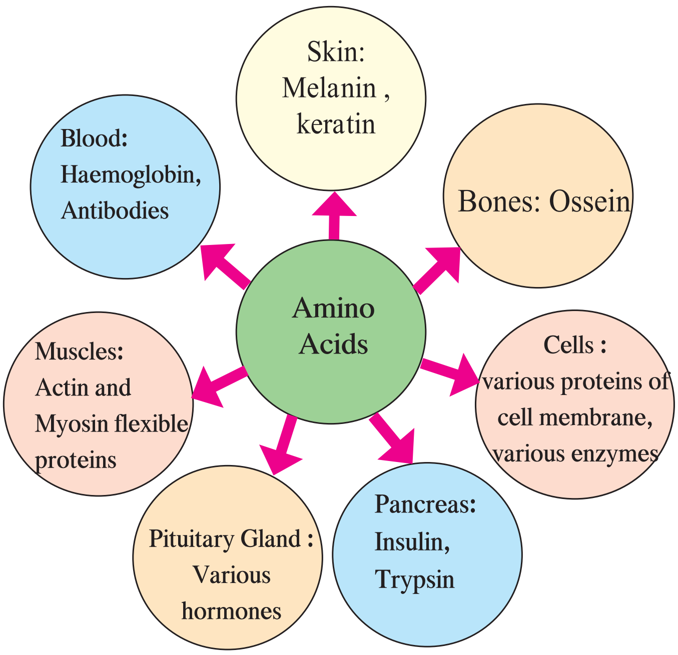
Can you tell?

1. Which type of cellular respiration performs complete oxidation of glucose?
2. Which cell organelle is necessary for complete oxidation of glucose?

Energy from different food components

Excess of the carbohydrates are stored in liver and muscles in the form of glycogen. What is the source of proteins? What are they made up of?

Proteins are the macromolecules formed by bonding together many amino acids. Proteins of animal origin are called as 'first class' proteins. We get 4 Kcal of energy per gram of proteins. Amino acids are obtained after digestion of proteins. Those amino acids are absorbed in the body and transported up to each organ and cell via blood. From these amino acids, organs and cells produce various proteins necessary for themselves and the whole body. Those examples are given in the following diagram.



2.4 Proteins and different amino acids obtained

Always remember

Excess of amino acids obtained from proteins are not stored in the body. They are broken down and the ammonia formed is eliminated out of the body. If necessary, excess of proteins are converted into other useful substances like glucose through the process of *gluconeogenesis.*

Plants produce the necessary amino acids from minerals *denovo* and thereby produce different proteins. An enzyme RUBISCO present in the plant chloroplasts is most abundant protein found in nature.

Can you recall?

From where do we obtain the lipids?

The substances formed by specific chemical bond between fatty acids and alcohol are called as lipids. Digestion of lipids consumed by us is nothing but their conversion into fatty acids and alcohol. Fatty acids are absorbed up and distributed everywhere within the body. From those fatty acids, different cells produce various substances necessary to themselves. Ex. the molecules called as phospholipids which are essential for producing plasma membrane are formed from fatty acids. Besides, fatty acids are used for producing hormones like progesterone, estrogen, testosterone, aldosterone, etc. and the covering around the axons of nerve cells. We get 9 KCal of energy per gram of lipids. Excess of lipids are stored in adipose connective tissue in the body.

Think:

1. Many times, you cannot eat hot food due to inflammation / ulceration in mouth.
2. Some persons experience difficulty in night vision since their childhood or adolescence.

Vitamins are a group of heterogeneous compounds of which, each is essential for proper operation of various processes in the body. There are main six types of vitamins, e.g. A, B, C, D, E and K. Out of these, A, D, E and K are fat-soluble whereas B and C are water-soluble. We have seen that, $FADH_2$ and $NADH_2$ are produced in the processes like glycolysis and Krebs cycle. Vitamins like riboflavin (Vitamin B_2) and nicotinamide (Vitamin B_3) respectively are necessary for their production.

Use your brain power

1. Many times, we experience dryness in mouth.
2. Oral rehydration solution (Salt-sugar-water) is frequently given to persons experiencing loose motions.
3. We sweat during summer and heavy exercise.

There is about 65 – 70% water in our body. Each cell contains 70% water weight by weight. Blood-plasma also contains 90% of water. Functioning of cells and thereby whole body disturbs even if there is a little loss of water from the body. Hence, water is an essential nutrient.
Along with all above mentioned nutrients, fibers are also essential nutrients. In fact, we cannot digest the fibers. However, they help in the digestion of other substances and egestion of undigested substances. We obtain the fibers from leafy vegetables, fruits, cereals, etc.

Internet is my friend

Collect information

1. What are symptoms of diseases like night blindness, rickets, beriberi, neuritis, pellagra, anaemia, scurvy?
2. What do you mean by coenzymes?
3. Find the full forms of FAD, FMN, NAD, NADP.
4. How much quantity of each vitamin is required every day?

Cell Division: An Essential Life Process

Can you tell?

1. What happens to the cells of injured tissue?
2. Whether new cells are formed during healing of wound?
3. Do the plants get injured when do we pluck the flowers? How are those wounds healed?
4. How does the growth of any living organism occur? Does the number of cells in their body increase? If yes, how?
5. How the new individual of a species is formed from existing one of same species?

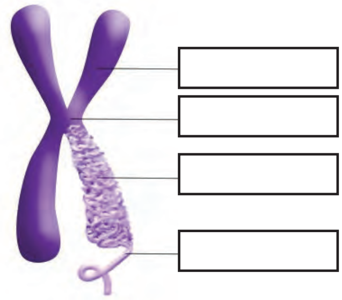
Cell division is one of the very important properties of cells and living organisms. Due to this property only, a new organism is formed from existing one, a multicellular organism grows up and emaciated body can be restored.

There are two types of cell division as mitosis and meiosis. Mitosis occurs in somatic cells and stem cells of the body whereas meiosis occurs in germ cells. Before study of cell division, we should know the structural organization of cell that we have studied earlier. Each cell has a nucleus. Besides, other cell organelles are also present. Let us study the cell division with the help of this information.

Before any type of cell division, the cell doubles up its chromosome number present in its nucleus i.e. if chromosome number is 2n, it is doubled up to 4n.

Can you recall? What is the shape of chromosome? Give its names in the figure.

A pair of each type of chromosome is present in 2n condition whereas single chromosome of each type is present in n condition and their structure is like the one shown in figure given beside.



2.5 Chromosome

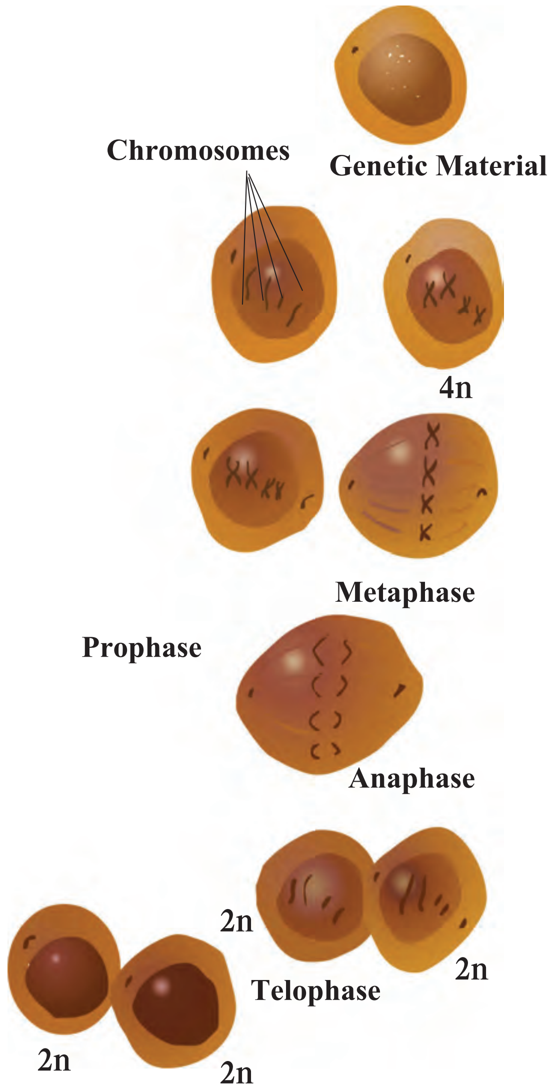
Mitosis

Somatic cells and stem cells divide by mitosis. Mitosis is completed through two main steps. Those two steps are karyokinesis (nuclear division) and cytokinesis (cytoplasmic division). Karyokinesis is completed through four steps.

A. Prophase : In prophase, condensation of basically thin thread-like chromosomes starts. Due to this, they become short and thick and they start to appear along with their pairs of sister chromatids. Centrioles duplicate and each centriole moves to opposite poles of the cells. Nuclear membrane and nucleolus start to disappear.

B. Metaphase : Nuclear membrane completely disappears in metaphase. Chromosomes complete their condensation and become clearly visible along with their sister chromatids. All chromosomes are arranged parallel to equatorial plane (central plane) of the cell. Special type of flexible protein fibers (spindle fibers) are formed between centromere of each chromosome and both centrioles.

C. Anaphase : In anaphase, centromeres split and thereby sister chromatids of each chromosome separate and they are pulled apart in opposite directions with the help of spindle fibers. Separated sister chromatids are called as daughter chromosomes. Chromosomes being pulled appear like bunch of bananas. In this way, each set of chromosomes reach at two opposite poles of the cell.

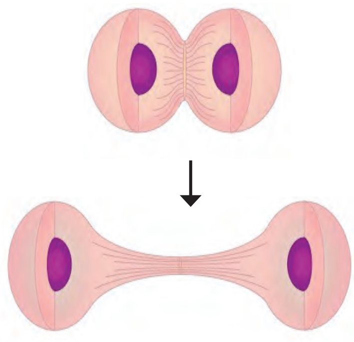


2.6 Mitosis

D. Telophase : The chromosomes which have reached at opposite poles of the cell now start to decondense due to which they again become thread-like thin and invisible. Nuclear membrane is formed around each set of chromosomes reached at poles. Thus, two daughter nuclei are formed in a cell. Nucleolus also appears in each daughter nucleus. Spindle fibers completely disappear.

In this way, karyokinesis completes and cytokinesis begins.

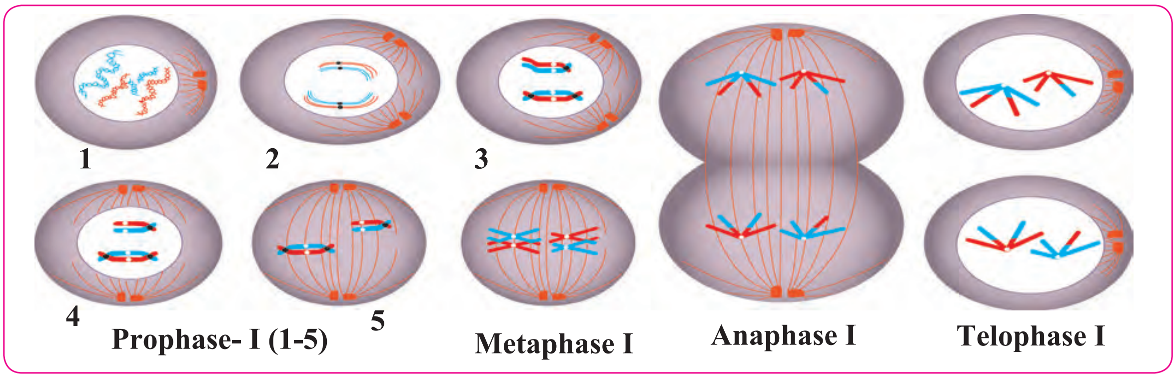
The cytoplasm divides by cytokinesis and two new cells are formed which are called as daughter cells. In this process, a notch is formed at the equatorial plane of the cell which deepens gradually and thereby two new cells are formed. However, in case of plant cells, instead of the notch, a cell plate is formed exactly along midline of the cell and thus cytokinesis is completed.



2.7 Cytokinesis

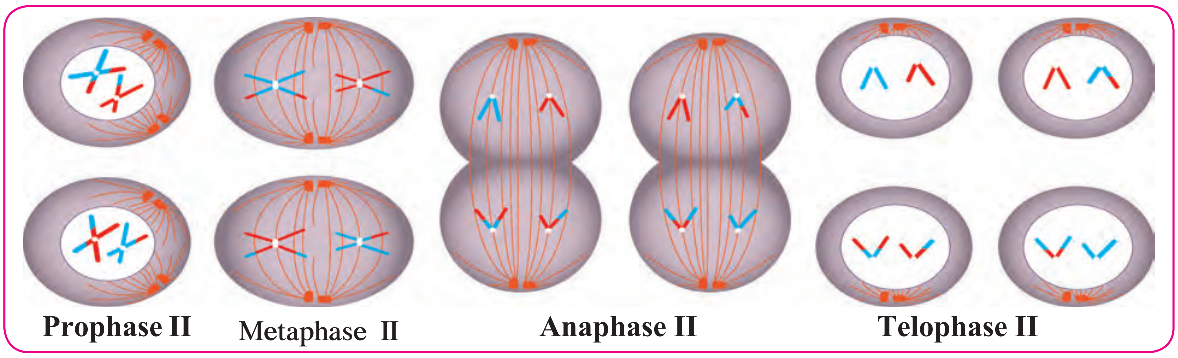
Mitosis is essential for growth of the body. Besides, it is necessary for restoration of emaciated body, wound healing, formation of blood cells, etc.

Meiosis:



2.8 Meiosis Part-I

Meiosis is completed through two stages. Those two stages are meiosis-I and meiosis-II. In meiosis-I, recombination / crossing over occur between homologous chromosomes and thereafter those homologous chromosomes (Not sister chromatids) are divided into two groups and thus two haploid cells are formed.



2.9 Meiosis Part -II

Meiosis-II is just like mitosis. In this stage, the two haploid daughter cells formed in meiosis-I undergo division by separation of recombined sister chromatids and four haploid daughter cells are formed. Process of gamete production and spore formation occurs by meiosis. In this type of cell division, four haploid (n) daughter cells are formed from one diploid (2n) cell. During this cell division, crossing over occurs between the homologous chromosomes and thereby genetic recombination occurs. Due to this, all the four daughter cells are genetically different from parent cell and from each other too.

Try this

Apparatus : Conical flask, glass slides, cover slips, forceps, compound microscope, watch glass, etc.

Materials : a medium sized onion , iodine solution, etc.

Procedure : Take a medium sized onion. Keep it in a conical flask filled with water in such a way that the roots of onion will be in contact with water. Observe the roots of onion after 4 – 5 days. Cut the tips of some of the roots and put them in a watch glass. Pour some drops of iodine in watch glass. Take one of the root tip on glass slide press it with the help of forceps. Add 1 – 2 drops of water and carefully place cover slip over it in such a way that air will not be trapped between. Observe the prepared glass slide under the compound microscope. Which phase of cell division did you observe? Sketch its figure.
Various phases of cell division occurring in root tips of onion are shown in the following figure. Which one of those could you see in the slide?

Use your brain power

1. What do you mean by diploid (2n) cell?
2. What do you mean by haploid (n) cell?
3. What do you mean by homologous chromosomes?
4. Whether the gametes are diploid or haploid? Why?
5. How are the haploid cells formed?
6. What is the importance of haploid cells?

2.10 Phases of mitosis in onion root tip

Use of ICT

Collect videos and photographs of different life processes in living organisms. Prepare a presentation and present it on the occasion of science exhibition

Books are my friend

Read different Encyclopaedias of technical terms in biology and anatomy and other reference books.

Exercise

1. Fill in the blanks and explain the statements.

a. After complete oxidation of a glucose molecules, ---- --- --- number of ATP molecules are formed.

b. At the end of glycolysis, ---- --- -- molecules are obtained.

c. Genetic recombination occurs in -- -- -- phase of prophase of meiosis-I.

d. All chromosomes are arranged parallel to equatorial plane of cell in -- -- -- phase of mitosis.

e. For formation of plasma membrane, --- --- --- molecules are necessary.

f. Our muscle cells perform -- -- -- type of respiration during exercise.

2. Write definitions.

a. Nutrition. b. Nutrients

c. Proteins. d. Cellular respiration

e. Aerobic respiration.

f. Glycolysis.

3. Distinguish between

a. Glycolysis and TCA cycle.

b. Mitosis and meiosis.

c. Aerobic and anaerobic respiration.

4. Give scientific reasons.

a. Oxygen is necessary for complete oxidation of glucose.

b. Fibers are one of the important nutrients.

c. Cell division is one of the important properties of cells and organisms.

d. Sometimes, higher plants and animals too perform anaerobic respiration.

e. Krebs cycle is also known as citric acid cycle.

5. Answer in detail.

a. Explain the glycolysis in detail.

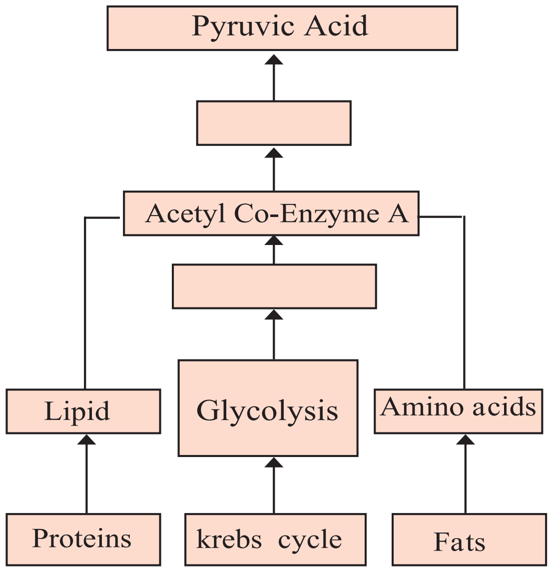
b. With the help of suitable diagrams, explain the mitosis in detail.

c. With the help of suitable diagrams, explain the five stages of prophase-I of meiosis.

d. How all the life processes contribute to the growth and development of the body?

e. Explain the Krebs cycle with reaction.

5. How energy is formed from oxidation of carbohydrates, fats and proteins? Correct the diagram given below.



Project :

With the help of information collected from internet, prepare the slides of various stages of mitosis and observe under the compound microscope.

KLQ2G7

3. Life Processes in Living Organisms Part – 2

- **Reproduction: Asexual and Sexual reproduction.**
- **Reproduction and modern technology**
- **Reproductive health**
- **Population Explosion**

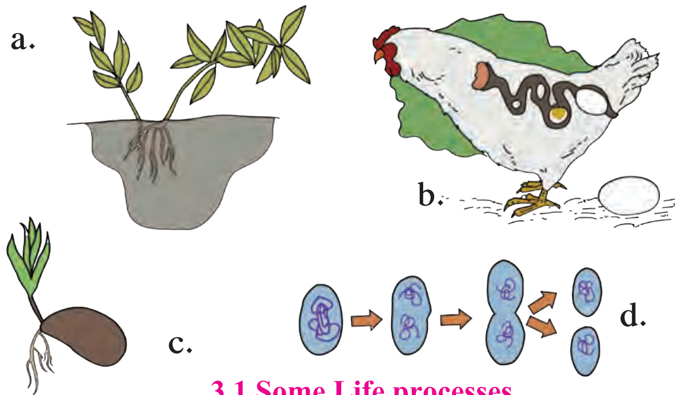
Can you recall?

1. Which are the important life processes in living organisms?
2. Which life processes are essential for production of energy required by body?
3. Which are main types of cell-division? What are the differences?
4. What is the role of chromosomes in cell-division?

We have studied various life processes in previous classes. All those life processes i.e. nutrition, respiration, excretion, sensation & response (control & co-ordination), etc. are essential to each living organism to remain alive. Besides these life processes, one more life process occurs in living organisms; it is reproduction. However, reproduction does not help the organism to remain alive but it helps to maintain the continuity of the species of that organism.

Observe

Observe the pictures and tell the life process which you identified.



3.1 Some Life processes

Can you tell?

1. What do we mean by maintenance of species?
2. Whether the new organism is genetically exactly similar to earlier one that has produced it?
3. Who determines whether the two organism of a species will be exactly similar or not?
4. What is the relationship between the cell division and formation of new organism of same species by earlier existing organism?

Formation of new organism of same species by earlier existing organism is called as reproduction. Reproduction is one of the various important characters of living organisms. It is also one of the various reasons responsible for evolution of each species. In living organisms, reproduction occurs mainly by two methods. Those two methods are- asexual and sexual reproduction.

Asexual reproduction

Process of formation of new organism by an organism of same species without involvement of gametes is called as asexual reproduction. As this reproduction does not involve union of two different gametes, the new organism has exact genetic similarity with the reproducing organism. This is uniparental reproduction and it occurs by mitotic cell division. Absence of genetic recombination is a drawback whereas fast process is advantage of this reproductive method.

A. Asexual reproduction in unicellular organisms

1. Binary Fission

Try this

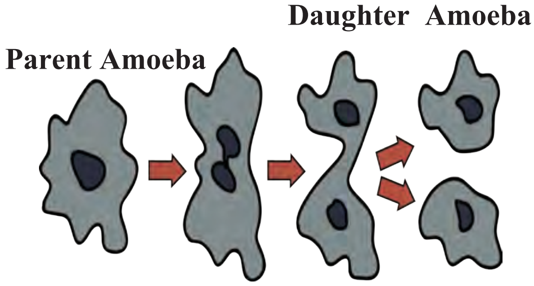
Activity 1 : Take a conical flask and collect the water in it from a pond having stagnant water and aquatic plants. Add some wheat grains and aquatic plants to it. Keep it for 3 – 4 days so that wheat grains & plants will decompose. Early in the morning on fourth day, take a glass slide and put a drop of that water over it. Carefully, put a cover-slip on that drop and observe under compound microscope.

You will be able to see many paramecia performing the binary fission.

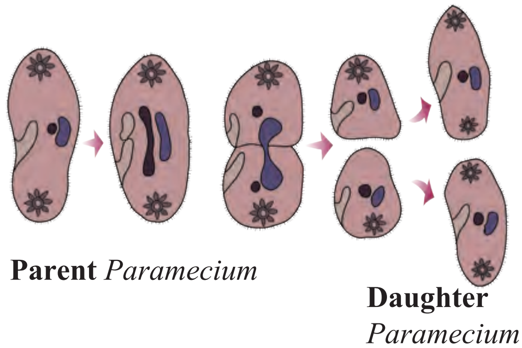
Prokaryotes (Bacteria), Protists (*Amoeba*, *Paramecium*, *Euglena*, etc.) and eukaryotic cell-organelle like mitochondria and chloroplasts perform asexual reproduction by binary fission. In this process, the parent cell divides to form two similar daughter cells. Binary fission occurs either by mitosis or amitosis.

Axis of fission / division is different in different protists. Ex.: *Amoeba* divides in any plane due to lack of specific shape; hence it is called as 'simple binary fission'. *Paramecium* divides by 'transverse binary fission' whereas *Euglena* by 'longitudinal binary fission'.

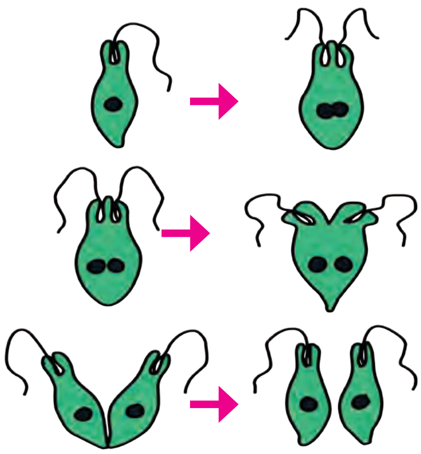
Binary fission is usually performed by living organisms during favourable conditions i.e. availability of abundant food material.



3.2 Simple binary fission: *Amoeba*



3.3 Transverse binary fission : Paramecium

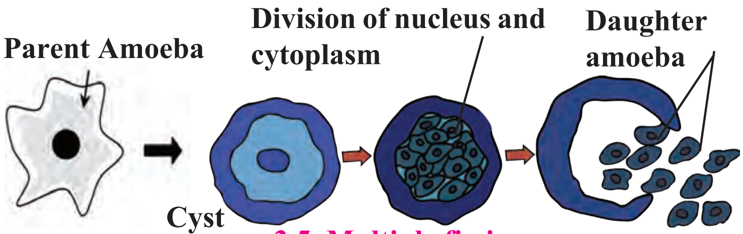


3.4 Longitudinal binary fission: *Euglena*

2. Multiple Fission

Asexual reproduction by multiple fission is performed by Amoeba and other similar protists. Amoeba stops the formation of pseudopodia and thereby movements whenever there is lack of food or any other type of adverse condition. It becomes rounded and forms protective covering around plasma membrane. Such encysted Amoeba or any other protist is called as 'Cyst'.

Many nuclei are formed by repeated nuclear divisions in the cyst. It is followed by cytoplasmic division and thus, many amoebulae are formed. They remain encysted till there are adverse conditions. Cyst breaks open on arrival of favourable conditions and many amoebulae are released.



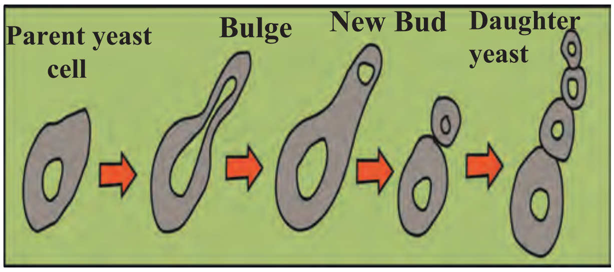
3.5 Multiple fission

Use your brain power

Does the parent cell exist after asexual reproduction- fission?

3. Budding:

Activity 2 : Bring the active dry yeast powder from market. Take 50 ml lukewarm water in a conical flask. Add 5 gm of active dry yeast powder and 10 gm table sugar to that water and mix well the mixture. Keep the flask in warm place and after an hour take a drop of that mixture on a clean glass slide. Put a cover-glass on that drop and observe it under the compound microscope.



3.6 Budding

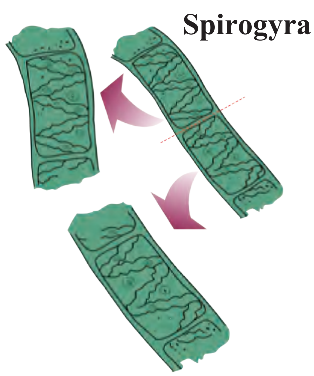
You will see the yeast cells performing budding i.e. a small bud coming out of many parent cells. Asexual reproduction occurs by budding in yeast- a unicellular fungus. Yeast cell produces two daughter nuclei by mitotic division, so as to reproduce by budding. This yeast cell is called as parent cell. A small bulge appears on the surface of parent cell. This bulge is actually a bud. One of the two daughter nuclei enters this bud. After sufficient growth, bud separates from the parent cell and starts to live independently as a daughter yeast cell.

B. Asexual reproduction in Multicellular organisms

1. Fragmentation:

This type of asexual reproduction occurs in multicellular organisms. In this type of reproduction, the body of parent organism breaks up into many fragments and each fragment starts to live as an independent new organism. This type of reproduction occurs in algae like *Spirogyra*, and sponges like *Sycon*.

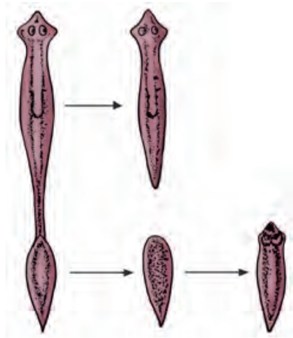
Whenever there is plenty of water and nutrients are available to *Spirogyra*, its filaments grow up very fast and break up into many small fragments. Each fragment starts to live independently as a new *Spirogyra* fiber.If the body of *Sycon* breaks up accidentally into many fragments, each fragment develops into new *Sycon*.



3.7 Fragmentation

2. Regeneration

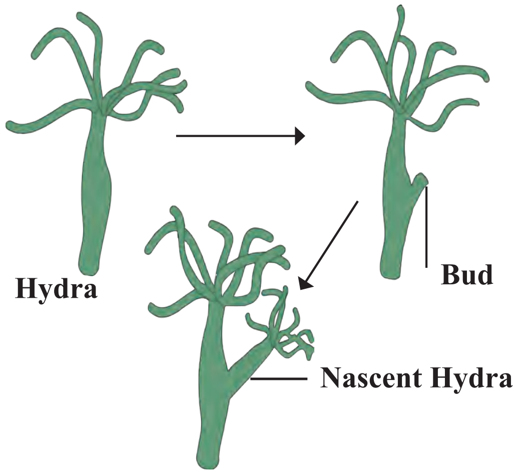
You may know that the wall lizard breaks up and discards some part of its tail in emergency. Discarded part is regenerated after a period. This is an example of limited regeneration. However, under certain situations, an animal- Planaria breaks up its body into two parts and thereafter each part regenerates remaining part of the body and thus two new Planaria are formed. This is called as regeneration.



3.8 Regeneration

3. Budding

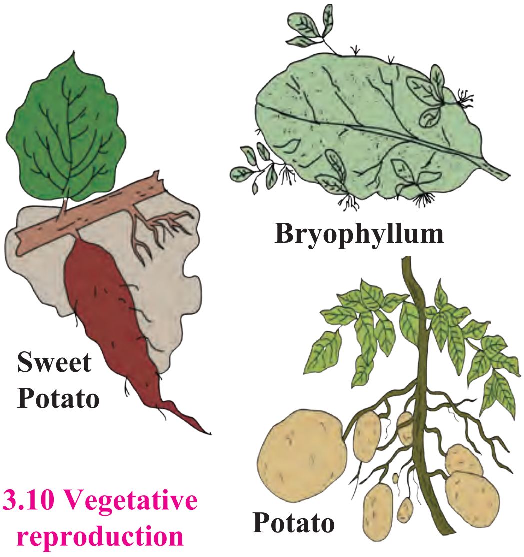
In case of *Hydra*, under favourable conditions, at specific part of its body, an outgrowth is formed by repeated divisions of regenerative cells of body wall. This outgrowth is called as bud. Bud grows up progressively and finally forms a small hydra. Dermal layers and digestive cavity of the budding hydra are in continuity with those of parent hydra. Parent hydra supplies nutrition to the budding hydra. Budding hydra separates from parent hydra and starts to lead an independent life when it grows up and becomes able to lead an independent life.



3.9 Budding

4. Vegetative Propagation

Reproduction in plants with the help of vegetative parts like root, stem, leaf and bud is called as vegetative reproduction. Vegetative propagation in potatoes is performed with the help of 'eyes' present on tuber whereas in *Bryophyllum* it is performed with the help of buds present on leaf margin. In case of plants like sugarcane & grasses, vegetative propagation occurs with the help of buds present on nodes.



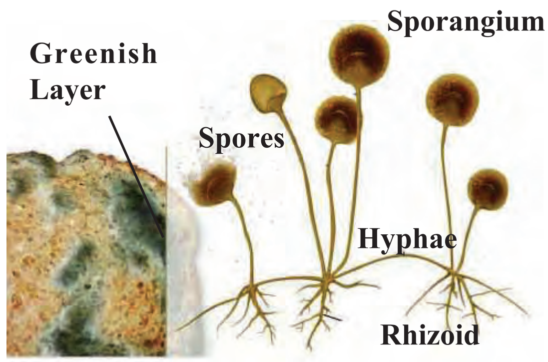
3.10 Vegetative reproduction

Try this

5. Spore Formation

Take a piece of wet bread or 'bhakari' and keep it in humid place. Fungus will grow on it within 2 – 3 days. Observe the fungus under compound microscope and draw its diagram.

Fungi like *Mucor* have filamentous body. They have sporangia. Once the spores are formed, sporangia burst and spores are released. Spores germinate in moist and warm place and new fungal colony is formed.



3.11 Spore formation

Sexual Reproduction

Sexual reproduction always occurs with the help of two germ cells. Female gamete and male gamete are those two germ cells. Two main processes occur in the sexual reproduction.

1. Gamete formation: Gametes are formed by the meiosis. In meiosis, chromosome number is reduced to half; hence haploid gametes are formed.

2. **Fertilization:** A diploid zygote is formed in this process by union of haploid male and female gametes. The zygote divides by mitosis and embryo is formed. The embryo develops to form new individual.

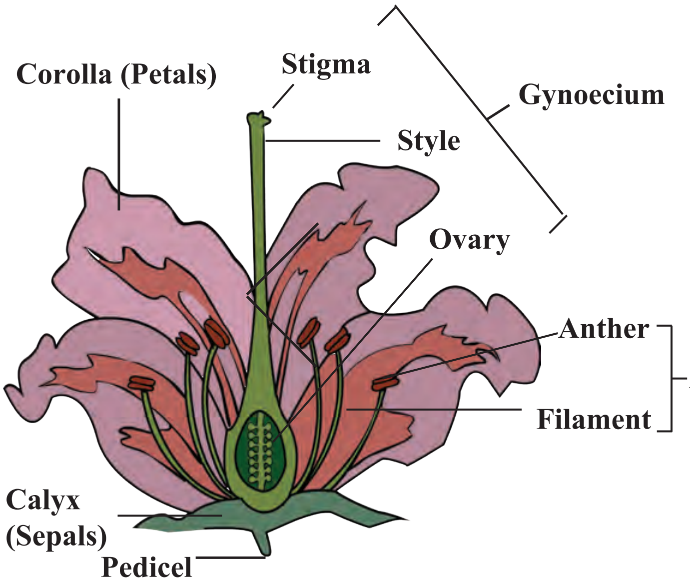
Two parents i.e. male parent and female parent are involved in this type of reproduction. Fusion of male gamete of male parent and female gamete of female parent occurs. Due to this, new individual always has the recombined genes of both the parents. Hence, the new individual shows similarities with the parents for some characters and has some characters different than both parents. Diversity in living organisms occurs due to genetic variation. Genetic variation helps the organisms to adjust with the changing environment and thereby to maintain their existence. Due to this, plants and animals can save themselves from being extinct.


Let's Think

1. What would have been happened if the male and female gametes had been diploid?
2. What would have been happened if any of the cells in nature had not been divided by meiosis?

A. Sexual reproduction in plants

Flower is structural unit of sexual reproduction in plants. It consists of four floral whorls as calyx, corolla, androecium and gynaecium; arranged in sequence from outside to inside. Androecium and gynoecium are called 'essential whorls' because they perform the function of reproduction whereas calyx and corolla are called as 'accessory whorls' because they are responsible for protection of inner whorls. Members of calyx are called as 'sepals' and they are green coloured. Members of corolla are called as 'petals' and they are variously colored.



3.12 Parts of flower

Female flower

Male flower

3.13 Papaya Flower

A flower is called as 'bisexual' if both whorls i.e. androecium and gynoecium are present in the same flower. Ex. *Hibiscus*. A flower is called as 'unisexual' if any one of the abovementioned two whorls is present in the flower. If only androecium is present, it is 'male flower' and if only gynoecium is present, flower is 'female flower'. Ex. Papaya.

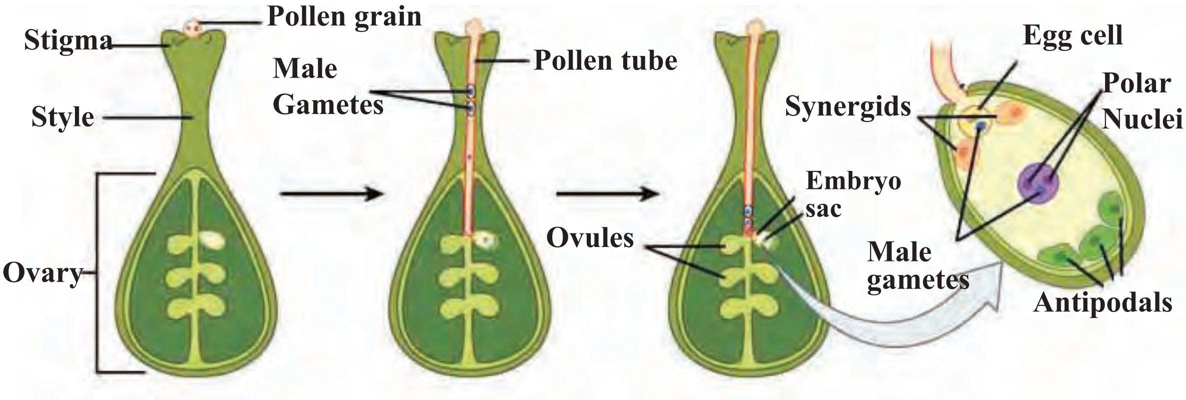
Many flowers have the stalk for support, called as 'pedicel' and such flowers are called as 'pedicellate' whereas flower without stalk is called as 'sessile'.

Androecium is male whorl and its members are called as stamens. Gynaecium is female whorl and its members are called as carpels.

These may be separate or united. Ovary is present at the basal end of each carpel. A hollow 'style' comes up from the ovary. Stigma is present at the tip of style. Ovary contains one or many ovules. Embryo sac is formed in each ovule by meiosis. Each embryo sac consists of a haploid egg cell and two haploid polar nuclei.

Pollen grains from anther are transferred to the stigma. This is called as pollination.

Pollination occurs with the help of abiotic agents (wind, water) and biotic agents (insects and other animals). Stigma becomes sticky during pollination Pollens germinate when they fall upon such sticky stigma i.e. a long pollen tube and two male gametes are formed. The pollen tube carries male gametes. Pollen tube reaches the embryo sac via style. Tip of the pollen tube bursts and two male gametes are released in embryo sac. One male gamete unites with the egg cell to form zygote. This is fertilization. Second male gamete unites with two polar nuclei and endosperm is formed. As two male nuclei participate in this process, it is called as double fertilization.



3.14 Double fertilization in angiosperms

Do you know?

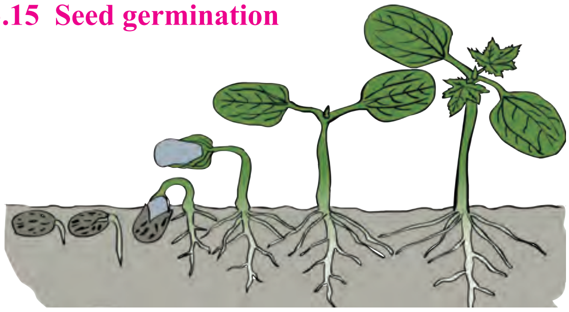
When pollination involves only one flower or two flowers borne on same plant, it is called as self-pollination whereas if it involves two flowers borne on two plants of same species, it is cross-pollination. While discovering the new high yielding and resistant varieties of plants, scientists bring about the pollination with the help of brush.

Use of ICT	Make an video album of pollination and show it in the class.

Try this

Take a suitable glass vessel like conical flask or beaker. Add some garden soil in it and sow some pulse grains in it in such a way that you can observe them through glass. Water it every day and record the changes.

3.15 Seed germination



Ovule develops into seed and ovary into fruit after fertilization. Seeds fall upon the ground when fruits break up and they germinate in the soil under favourable conditions. Zygote develops at the cost of food stored in endosperm of seed and thus a new plantlet is formed. This is called as seed germination.

B. Sexual reproduction in human being

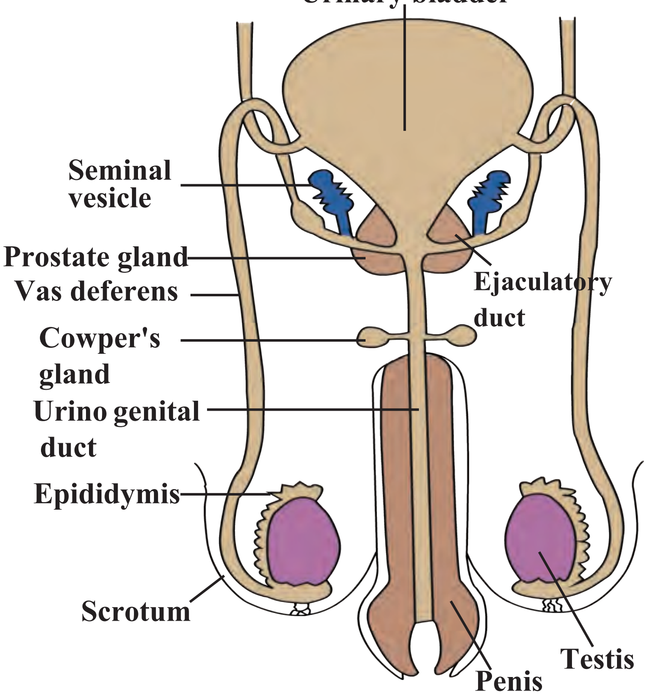
Can you recall?

1. Which different hormones control the functions of human reproductive system through chemical coordination?
2. Which hormones are responsible for changes in human body occurring during on set of sexual maturity?
3. Why has the Government of India enacted the law to fix the minimum age of marriage as 18 in girls and 21 in boys?

We have studied in the chapter of heredity and variation that men have XY sex-chromosomes and women have XX sex-chromosomes. Reproductive system with specific organs develops in the body of men and women due to these sex-chromosomes only. X-chromosome is present in men and women whereas Y-chromosome is present in men only. Now we shall study the structure and functions of human reproductive system.

Human male reproductive system

Male reproductive system of humans consists of testes, various ducts and glands. Testes are present in the scrotum, outside the abdominal cavity. Testes contain numerous seminiferous tubules. Germinal epithelium present in the tubules divide by meiosis to produce sperms. Those sperms are sent forward through various tubules. Sequence of those tubules is as-rete testis, vas eferens, epididymis, vas deferens, ejaculatory duct and urinogenital duct. As the sperms are pushed forwards from one duct to next, they become mature and able to fertilize the ovum.



3.16 Male reproductive system of human

Seminal vesicles secrete their secretion in ejaculatory ducts whereas prostate gland and Cowper's glands secrete their secretions in urinogenital duct. Semen is formed of sperms and secretions of all these glands. Semen is ejaculated out through penis. All the organs of male reproductive system are paired except urinogenital duct, prostate gland, penis & scrotum.

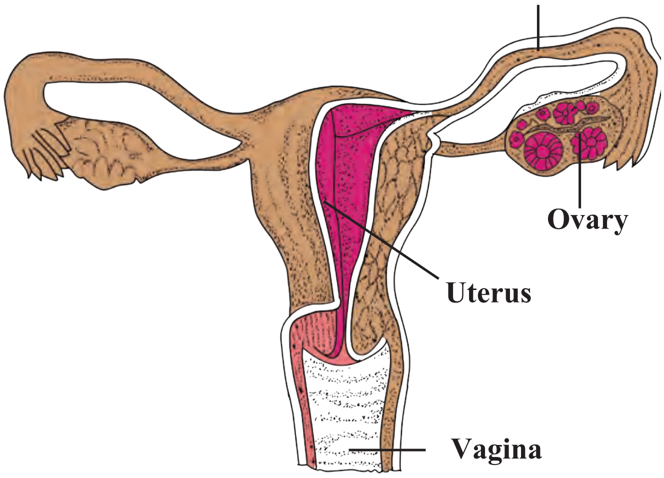
Surprising Facts

1. Length of each epididymis is about 6 meters.
2. Length of a sperm is about 60 micrometers.
3. Such a small sperm has to cross the distance of approximately 6.5 meter while passing out of male reproductive system.
4. Sperm needs large amount of energy. For this purpose, fructose is present in the semen.

Human female reproductive system

All organs of female reproductive system are in abdominal cavity. It includes a pair of ovaries, a pair of oviducts, single uterus and a vagina. Besides, a pair of Bartholin's gland is also present.

Generally, every month, an ovum is released in abdominal cavity alternately from each ovary. Free end of oviduct is funnel-like. An opening is present at the center of it. Oocyte enters the oviduct through that opening. Cilia are present on inner surface of oviduct. These cilia push the oocyte towards uterus.



3.17 Human female reproductive system

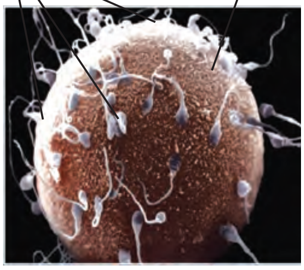
Gamete Formation

Both gametes i.e. sperm and ovum are formed by meiosis. Sperms are produced in testes of men from beginning of maturation (puberty) till death. However, in case of women, at the time of birth, there are 2 – 4 million immature oocytes in the ovaries of female foetus. An oocyte matures and is released from ovary every month from the beginning of maturity up to the age of menopause (approximately 45 years of age). Menopause is the stoppage of functioning of female reproductive system. At the age of about 45 – 50 years, secretion of hormones controlling the functions of female reproductive system either stops or becomes irregular. This causes the menopause.

Fertilization

Formation of zygote by union of sperm and ovum is called as fertilization. Fertilization is internal in humans. Semen is ejaculated in vagina during copulation. Sperms, in the numbers of few millions start their journey by the route of vagina – uterus – oviduct. One of those few million sperms fertilize the only ovum present in the oviduct.

From the age of puberty up the menopause (from 10 – 17 years of age up to 45 – 50 years), an ovum is released every month from the ovary. i.e. out of 2 – 4 million oocytes, approximately only 400 oocytes are released up to the age of menopause. Remaining oocytes undergo degeneration.

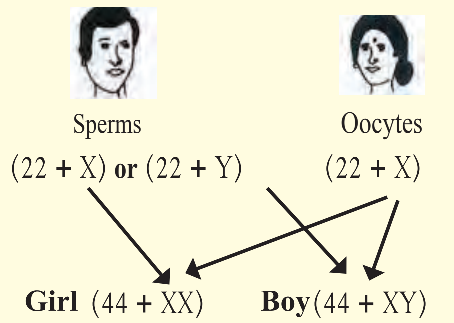


3.18 Fertilization

Oocytes released from ovaries during last few months nearing the age of menopause are 40 – 50 years old. Their ability of division has been diminished till now. Due to this, they cannot complete meiotic division properly. If such oocytes are fertilized, the new-borns produced from them may be with some abnormalities like Down's syndrome.

Do you know?

1. The chromosome number in germ cells producing the gametes are diploid i.e. 2n. It includes 22 pairs of autosomes and 1 pair of sex-chromosomes i.e. (44 + XX or 44 + XY). These germ cells divide by meiosis. Due to this, gametes contain only haploid (n) number of chromosomes i.e. (22 + X or 22 + Y). Two types of sperms are produced as (22 + X) or (22 + Y) whereas oocytes are produced of only one type as (22 + X).

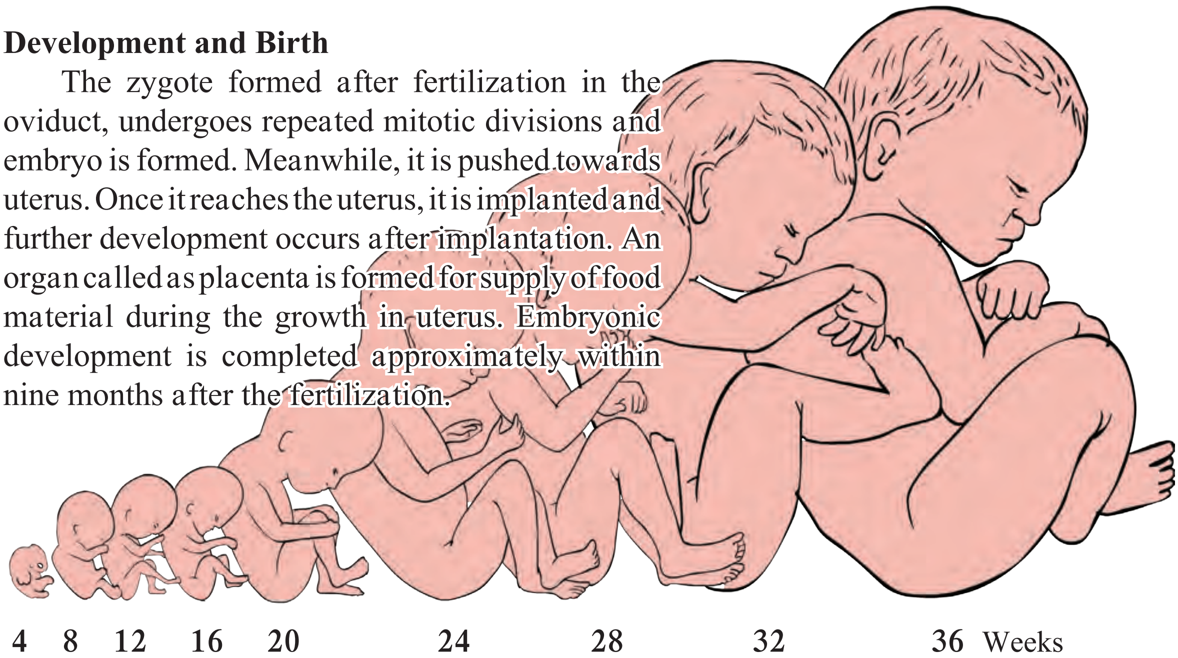


3.19 Sex determination in human being

2. Both, sperms and oocytes are produced by meiosis. In case of sperms, process of meiotic division is completed before the sperms leave male reproductive tract. However, in case of oocytes, process of meiotic division completes after ovulation; during fertilization in oviduct.

Development and Birth

The zygote formed after fertilization in the oviduct, undergoes repeated mitotic divisions and embryo is formed. Meanwhile, it is pushed towards uterus. Once it reaches the uterus, it is implanted and further development occurs after implantation. An organ called as placenta is formed for supply of food material during the growth in uterus. Embryonic development is completed approximately within nine months after the fertilization.



3.20 Growth of an embryo

Always remember

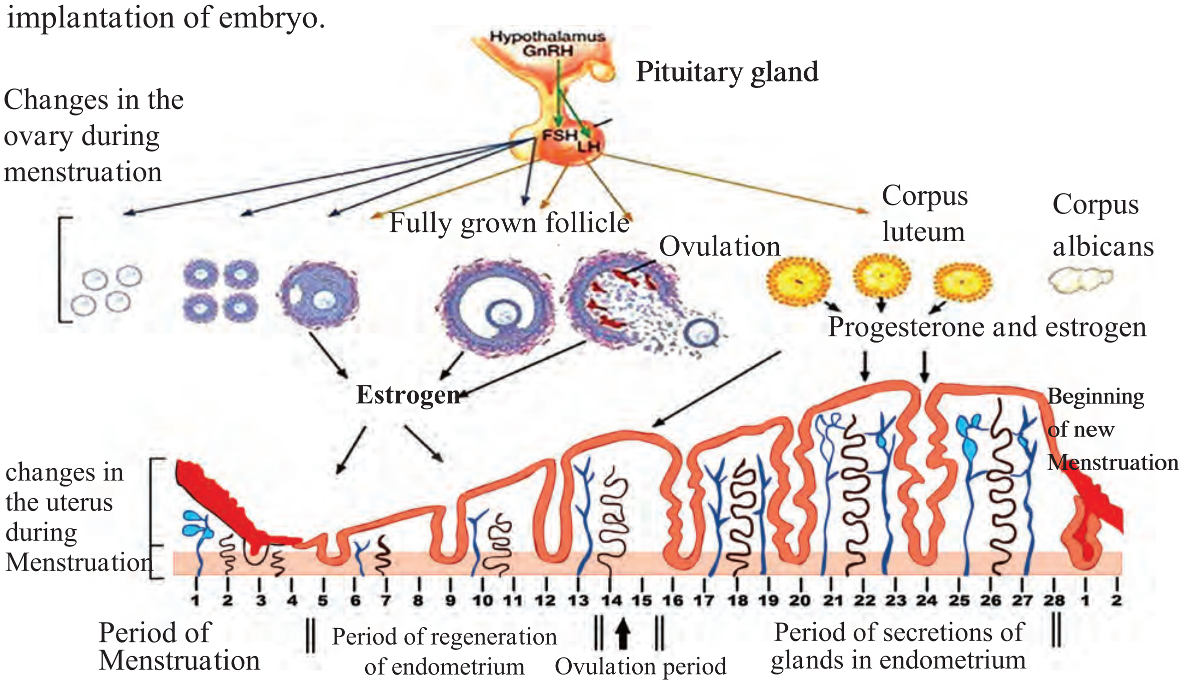
The man is totally responsible, whether the couple will have a boy or a girl child. During zygote formation, man contributes either X or Y chromosome to the next generation, but woman contributes only X-sex chromosome to the next generation. At the time of fertilization, if X- chromosomes comes from male, the child will be a girl and if Y-chromosome comes then the child will be a boy. **Thinking of this, is it right to consider the mother responsible for a girl child? We all must take efforts to stop female foeticide.**

Can you tell?

1. Which hormone is released from pituitary of mother once the foetal development is completed?
2. Under the effect of that hormone, which organ of the female reproductive system starts to contract and thereby birth process (Parturition) is facilitated?

Menstrual Cycle:

Female reproductive system undergoes some changes at puberty and those changes repeat at the interval of every 28 – 30 days. These repetitive changes are called as menstrual cycle. Menstrual cycle is a natural process, controlled by four hormones. Those four hormones are follicle stimulating hormone (FSH), luteinizing hormone (LH), estrogen and progesterone. One of the several follicles in the ovary starts to develop along with the oocyte present in it, under the effect of follicle stimulating hormone. This developing follicle secretes estrogen. Endometrium of the uterus starts to develop (during first cycle) or regenerate (during subsequent cycles) under the effect of estrogen. Meanwhile, developing follicle completes its development. It bursts under the effect of luteinizing hormone and oocyte is released. This is called as ovulation. Remaining tissue of the burst follicle forms the corpus luteum. Corpus luteum starts to secrete progesterone. Endometrial glands secrete their secretion under the effect of progesterone. Such endometrium is ready for implantation of embryo.



3.21 Menstrual cycle

If oocyte is not fertilized within 24 hours, corpus luteum becomes inactive and transforms into corpus albicans. Due to this, secretion of estrogen and progesterone stops completely. Endometrium starts to degenerate in absence of these two hormones. Tissues of degenerating endometrium and unfertilized ovum are discarded out through vagina. This is accompanied with continuous bleeding. Bleeding continues approximately for five days. This is called as menstruation.

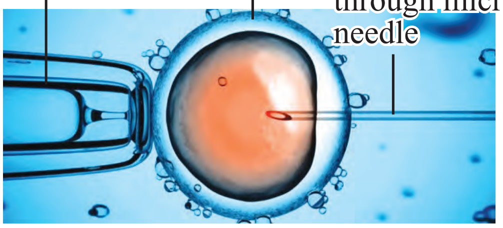
Unless the oocyte is fertilized and embryo is implanted, this process is repeated every month. If the embryo is implanted, repetition of this cycle is temporarily stopped till the parturition and thereafter period of breast feeding. Menstrual cycle is a natural process and the women experience severe pains during this period. Severe weakness is felt due to heavy bleeding. There is higher possibility of infections too during this overall period. Due to all such reasons, there is need of rest along with special personal hygiene.

Reproduction and Modern Technology

Many couples cannot have children due to various reasons. In case of women, irregularity in menstrual cycle, difficulties in oocyte production, obstacles in the oviduct, difficulties in implantation in uterus and many other reasons are responsible for this. Absence of sperms in the semen, slow movement of sperms, anomalies in the sperms are the reasons in case of males. But now with the help of advanced medical techniques like IVF, Surrogacy, Sperm bank the childless couples can have a child.

In Vitro Fertilization (IVF)

In this technique, fertilization is brought about in the test-tube and the embryo formed is implanted in uterus of woman at appropriate time. IVF technique is used for having the child in case of those childless couples who have problems like less sperm count, obstacles in oviduct, etc.



3.22 Fertilization in a test tube

Surrogacy

Some women have problems in implantation of embryo in uterus. Such women can take the help of the modern remedial technique called as surrogacy. In this technique, oocyte is collected from the ovary of the woman having problem in implantation in uterus. That oocyte is fertilized in test-tube with the help of sperms collected from her husband. The embryo formed from such fertilization is implanted in the uterus of some other woman having normal uterus. Such a woman, in whose uterus the embryo is implanted, is called as surrogate mother.

3.23 Surrogacy

Sperm Bank/ Semen Bank

There are various problems in sperm production as mentioned above, in case of many men. So as to have the children in case of such couples, new concept of sperm bank has been introduced. This concept is similar to blood bank. Semen ejaculated by the desired men is collected after their thorough physical and medical check-up and stored in the sperm bank.

As per the wish of needful couple, oocyte of woman of the concerned couple is fertilized by IVF technique using the semen from sperm bank. Resultant embryo is implanted in the uterus of same woman. Name of the semen donor is strictly kept secret as per the law.

Twins

Two embryos develop simultaneously in the same uterus and thus two offsprings are delivered simultaneously. Such offsprings are called as twins. Many couples have twins. There are two main types of twins as- monozygotic twins and dizygotic twins.

Monozygotic twins are formed from single embryo. During early period of embryonic development (within 8 days of zygote formation), cells of that embryo divide into two groups.

3.24 Twin girls: age 18 months

Those two groups develop as two separate embryos and thus monozygotic twins are formed. Such twins are genetically exactly similar to each other. Due to this, such twins are exactly similar in their appearance and their gender is also same i.e. both will be either boys or girls.

In case of monozygotic twins, if the embryonic cells are divided into two groups 8 days after the zygote formation; there is high possibility of formation of conjoined twins (Siamese twins). Such twins are born with some parts of body joined to each other. Some organs are common in such twins.

Occasionally, two oocytes are released from the ovary of woman and both oocytes are fertilized by two separate sperms and thus two zygotes are formed. Two embryos are formed from those two zygotes and both of those embryos are separately implanted in the uterus and thus dizygotic twins are delivered after complete development. Such twins are genetically different and may be same of different by gender.

Internet is my friend

You may have read that sometimes a woman may deliver more than two offsprings at a time. Collect more information from internet about reasons for such incidences.

Reproductive health

A person's state of being physical, mental and social strongness is called as health. In our country, there seems to be lack of awareness regarding reproductive health due to various reasons like social customs, traditions, illiteracy, shyness, etc. Especially, there seems to be indifference towards the reproductive health of women.

Occurrence of menstrual cycle is related with reproductive and overall health of women. Now a day, women are working at par with men. Due to this, they have to stay outdoors for whole day. Bleeding occurs during menstrual cycle. Due to this, private organs (genitals) need to be maintained clean time to time, otherwise, problems regarding reproductive health may arise. Some problems regarding reproductive health may arise in men too. It is essential to maintain the cleanliness of their genitals.

Among the various sexual diseases, syphilis and gonorrhoea occur on large scale. Both of these diseases are caused by bacteria. Occurrence of chancre (patches) on various parts of body including genitals, rash, fever, inflammation of joints, alopecia, etc. are the symptoms of syphilis. Painful and burning sensation during urination, oozing of pus through penis and vagina, inflammation of urinary tract, anus, throat, eyes, etc. are symptoms of gonorrhoea.

Do you know?

Population Explosion

Excessive growth of population within short duration is called as population explosion. You may have realized from the table given besides about fast population growth of India. We have to face various problems like unemployment, decreasing per capita income and increasing loan, stress on natural resources, etc. There is only one solution for all such problems and it is population control. Family planning is essential for this.

Year	Population
1901	238396327
1911	252093390
1921	251321213
1931	278977238
1941	318660580
1951	358142161
1961	439234771
1971	548159652
1981	683329097
1991	846421039
2001	1028610328
2011	1210854977

Get information

Visit a public health center nearby your place and collect the information through an interview of health officer about meaning and various methods of family planning.

Exercise

1. Complete the following chart.

Asexual reproduction	Sexual reproduction
1. Reproduction that occurs with the help of somatic cells is called as asexual reproduction.	1. ....
2	2. Male and female parent are necessary for sexual reproduction.
3. This reproduction occurs with the help of mitosis only.	3. ....
4.	4. New individual formed by this method is genetically different from parents.
5. Asexual reproduction occurs in different individuals by various methods like binary fission, multiple fission, budding, fragmentation, regeneration, vegetative propagation, spore production, etc.	5. ....

2. Fill in the blanks.

a. In humans, sperm production occurs in the organ ----------------.

b. In humans, ------ chromosome is responsible for maleness.

c. In male and female reproductive system of human, ------------- gland is same.

d. Implantation of embryo occurs in ------

e. ---------- type of reproduction occurs without fusion of gametes.

f. Body breaks up into several fragments and each fragment starts to live as a new individual. This is -- -- -- -- -- -- -- type of reproduction.

g. Pollen grains are formed by -- ----- -- -- -- division in locules of anthers.

3. Complete the paragraph with the help of words given in the bracket.

(Luteinizing hormone, endometrium of uterus, follicle stimulating hormone, estrogen, progesterone, corpus luteum) Growth of follicles present in the ovary occurs under the effect of ----- This follicle secretes estrogen. -- --- - -- -- -- grows / regenerates under the effect of estrogen. Under the effect of --- ---, fully grown up follicle bursts, ovulation occurs and -- --- -- -- is formed from remaining part of follicle. It secrets --- --- -- -- and --- --- -- -- --. Under the effect of these hormones, glands of -- -- -- -- are activated and it becomes ready for implantation.

4. Answer the following questions in short.

a. Explain with examples types of asexual reproduction in unicellular organism.

b. Explain the concept of IVF.

c. Which precautions will you follow to maintain the reproductive health?

d. What is menstrual cycle? Describe it in brief.

5. In case of sexual reproduction, new-born show similarities about characters. Explain this statement with suitable examples.

6. Sketch the labelled diagrams.

a. Human male reproductive system.

b. Human female reproductive system.

c. Flower with its sexual reproductive organs.

d. Menstrual cycle.

7. Give the names.

a. Hormones related with male reproductive system.

b. Hormones secreted by ovary of female reproductive system.

c. Types of twins.

d. Any two sexual diseases.

e. Methods of family planning.

8. Gender of child is determined by the male partner of couple. Explain with reasons whether this statement is true or false.

9. Explain asexual reproduction in plants.

10. Modern techniques like surrogate mother, sperm bank and IVF technique will help the human beings. Justify this statement.

11. Explain sexual reproduction in plants.

Activity :

1. Collect the official data about present and a decade old population of various Asian countries and plot a graph of that data. With the help of it, draw your conclusions about demographic changes.

2. With the help of your teacher, compose and present a road show to increase the awareness about prenatal gender detection and gender bias.

KLYXHU

4. Environmental management

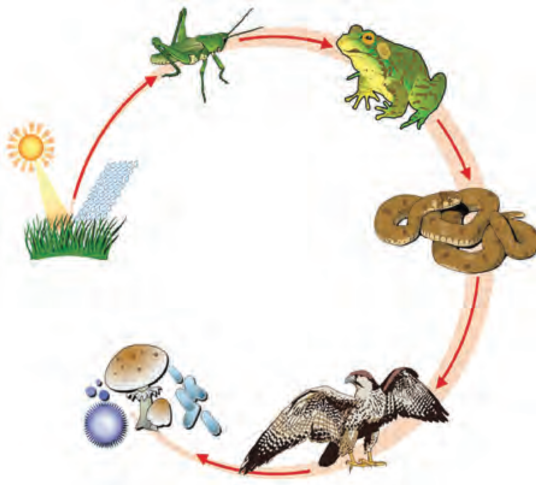
- Ecosystem – A review
- Environment Conservation
- Biodiversity hotspots
- Environment and Eco-system
- Environment management

Can you recall?

1. What is ecosystem? Which are its different components?
2. Which are the types of consumers? What are the criteria for their classification?
3. What may be the relationship between lake and birds on tree?
4. What is difference between food chain and food web?

Think and Answer

1. Write the name and category of each of the component shown in picture.
2. What is necessary to convert this picture into food web? Why?



4.1 Food chain

Ecosystem A review

Ecosystem is formed by biotic and abiotic factors and their interactions with each other. Each factor plays very important role in the ecosystem. Producers like plants are important. Herbivores like deer, goats, sheep, cattle, horses, camels, etc. feeding upon producers are also important. Predators like lion and tiger which prevent the overpopulation of herbivores are also equally important. A question may arise in our mind that whether the caterpillars found in nature, organisms present in filthy places, termites, insects present in dung, are really useful? However, those organisms are also important though they are dirty. They are responsible for cleaning the environment.

It means that our existence is due to these factors present around us. Hence, we should care for these factors.

Think:

If fallen foliage in forest, dead trees, and carcasses in and around villages had not been decomposed for years……

Discuss

'Jivo Jivasya Jivanam'

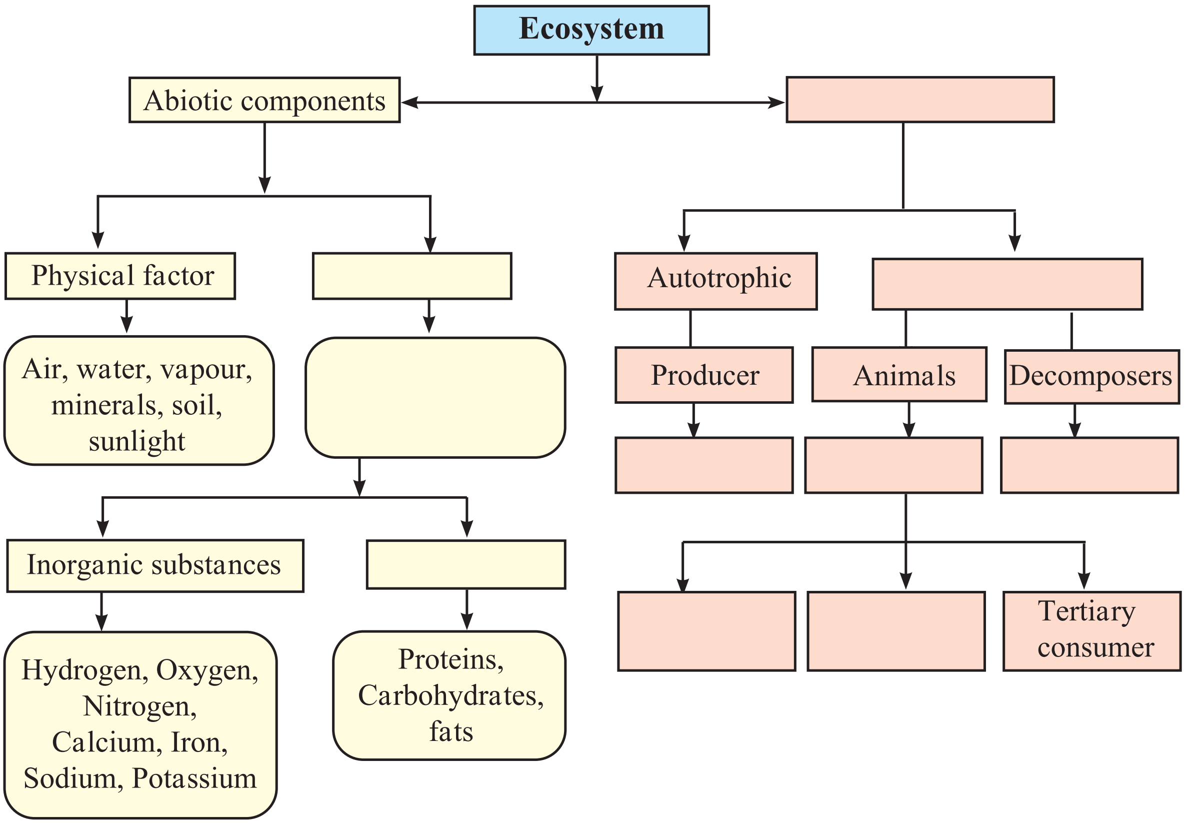
Can you recall?

1. Which are different trophic levels in food chain?
2. What is energy pyramid?

Let's Think

A bird building nest on a tree feeds upon the fishes in nearby pond. Whether this bird is part of both i.e. tree as well as pond ecosystem?

Complete the chart Fill up the blank boxes and display the completed chart in classroom.



Paddy is cultivated on large scale in various states of South India. Paddy fields are frequently attacked by grasshoppers. Similarly, frogs are also present in large number in the mud of paddy fields, to feed upon grasshoppers and snakes are also present therein to feed upon their favourite food- frogs.

However, if frog population declines all of a sudden,

Let's Think

1. What will be the effect on paddy crop?
2. Number of which consumers will decline and which will increase?
3. What will be overall effect on that ecosystem?

Can you tell?

1. What is environment?
2. What is included in environment?

Relationship between Environment and Ecosystem

Environment is a broad concept. Physical, chemical and biological factors affecting the living organisms in any possible way is collectively called as environment. In short, environment is the condition in surrounding. It includes many biotic, abiotic, natural and artificial factors. There are two main types of environment. One is natural environment and other is artificial environment.

Natural environment consists of air, atmosphere, water, land, living organisms, etc. Continuous interactions occur between biotic and abiotic factors. Their interactions are very important. Artificial environment is also affecting the natural environment directly or indirectly. Basically, environment consists of two basic factors- 1. Biotic factors, 2. Abiotic factors. The science that deals with the study of interactions between biotic and abiotic factors of the environment is called as ecology. Basic functional unit used to study the ecology is called as ecosystem.

Environment consists of many ecosystems. We have studied some ecosystems in earlier classes. A small pond is an ecosystem whereas the Earth is largest ecosystem. In brief, biotic and abiotic factors occupying a definite geographical area and their interactions collectively constitute the ecosystem.

Can you recall?

Which cycles are operated in environment? What is their importance?

Environmental balance is maintained through continuous operation of various natural cycles like water cycle, carbon cycle, gaseous cycles like nitrogen cycle, oxygen cycle, etc. Environmental balance is also maintained due to various food chains of ecosystem.

Human existence is totally impossible without the existence of nature. Hence, it is basic responsibility of human being to preserve the nature without disturbing its balance. It is said that we have got this Earth planet on lease from our future generations and not as an ancestral property from our ancestors. Hence we should not forget to conserve it for ourselves and for future generations.

Environmental Conservation

Can you tell?

1. Which factors affect the environment? How?
2. What will happen if number of consumers in environment goes on increasing gradually?
3. What will be the effect of industry established on river bank on the river ecosystem?

When some natural factors of environment and some artificial polluted factors harm the environment, it creates imbalance between various factors of the environment and ultimately affects the existence of biotic factors.

Observe and fill the information

Observe the environment around you. Complete the following flow chart.

Factors affecting environment

Natural factors

Earthquakes ....
....

Manmade factors

Deforestation.....................
....

"The Earth is sufficient to satisfy everyone's need but not the greed."
- Mahatma Gandhi

At present, many environmental problems have been arisen due to effect of various natural and artificial factors on the Earth. Environmental pollution is one of those main factors. Generally, contamination of any material is its pollution. Unnecessary and unacceptable change in the surrounding environment due to natural events or human activities are called as environmental pollution. i.e. direct or indirect changes in physical, chemical and biological properties of air, water and soil which will be harmful to humans and other living beings is environmental pollution. Various reasons like population explosion, fast industrialization, and indiscriminate use of natural resources, deforestation, and unplanned urbanization are responsible for environmental pollution.

Can you recall?

1. Which are the types of pollution?
2. What do we mean by natural and artificial pollution?

4.2 Fog and pollution in city- A problem

Pollution is a broad concept. Various types of pollutions like that of air, water, sound, soil, thermal, light, pollution occur around us. Ultimate adverse effect of all these is on existence of all the living organisms and out of this, environmental conservation has become the need of hour.

Complete the Chart

We have studied the air pollution, water pollution and soil pollution in detail in earlier classes. Based on that, complete the following chart.

	Air pollution	Water pollution	Soil pollution
Compo-nents	**Gases :** CO_2, CO, Hydro-carbons, Sulphur, NOx, Hydrogen sulphide, etc. **Solid** : dust, ash, carbon, lead, asbestos, etc.		
Source		Industrial wastes, Domestic waste, Sewage, Chemicals discharged from Industries, Pesticides used in agriculture.	
Effect			Soil erosion, Retarded growth of plants/ crops, Nutritional deficiency etc.
Control Measure			

Do you know?

Radioactive pollution: Radioactive pollution can occur due to two reasons as natural and artificial. UV and IR radiations are natural radiations whereas X-rays and radiations from atomic energy plants are artificial radiations. Chernobyl, Windscale, and Three Miles Island mishaps are the major mishaps of the world till present. Thousands of people have been affected for long term due to these accidents. Some of the effects of radiations are as follows-

1. Cancerous ulceration occurs due to higher radiations of X-rays.
2. Tissues in the body are destroyed.
3. Genetic changes occur.
4. Vision is adversely affected.

Use your brain power

Why is it said that pollution control is important?

Need of environmental conservation

General public is not aware about the rules of environment conservation. There should be large scale participation of the people in environment conservation. It will be possible to answer the environmental problems only if environmental protection-conservation becomes an effective public movement. For this purpose, values like positive attitude and affection towards environment, knowledge about it, etc. should be inculcated among the children since their childhood. This will help to make the future generations more aware about environmental conservation and protection. So as to achieve this, it is necessary to increase the awareness through education.

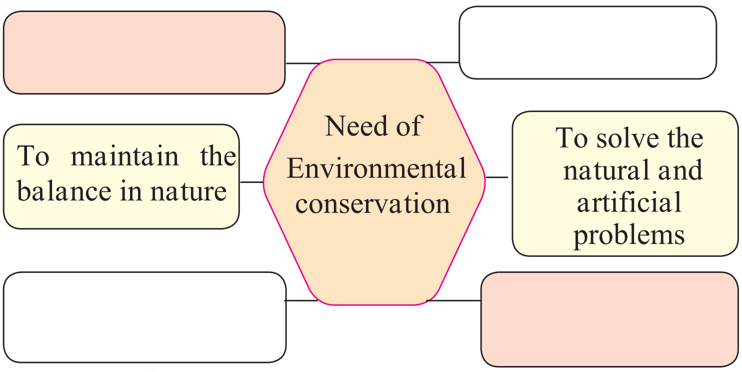
Today, all the developed, developing and underdeveloped countries have accepted the responsibility of environmental protection. Their efforts are in that direction. They have defined the future plans about environmental protection and have constituted the necessary laws.

A peek into the history

In 1972, United Nations Environment Program (UNEP) has been established in a conference arranged on human and environment in which environmental problems were discussed. Afterwards, in India, a separate environmental department had been established after through discussion on environmental problems. Ministry of environment and forests is involved in planning, inducting and increasing awareness about environment and forest through various programs since 1985.

Complete the Chart

Nowadays, we are observing the environmental degradation everywhere. Complete the flow chart given besides with the help of environment.



Environmental Conservation. Our social responsibility

Since existence of human, there is interrelationship between human and environment. Human stepped on the Earth long after formation of Earth. On the Earth, human being proved its superiority as compared to other animals with the help of characters like intelligence, memory, imaginary ability, etc. Human established domination over the nature. Human utilized all the natural resources as much as possible. In an attempt to live a satisfactory life, human kept on snatching from the nature as much as possible and this led to increase in problems. From this entire scenario, we can understand that human has crucial role in maintaining the environmental balance. If human has disturbed the environmental balance, then human itself only can conserve and improve the quality of nature. Many times, general public is unknown that its activities are harmful to environment and thereby unknowingly many activities happen.

Search How do butterflies contribute to environmental balance?

Do You Know?

Laws enacted about environmental conservation:

Forest (Conservation) Act, 1980.

The land reserved for forest conservation has been prohibited to use for any other purpose by this law. Ex. Permission of central government is compulsory for mining activities. Any person who disobeys this law is entitled to imprisonment for 15 days.

Environmental (Protection) Act, 1986.

Purpose of this act is to control the pollution and punish the persons or institutes harming the environment. Any person or factory is prohibited by this act from releasing the pollutants in atmosphere beyond a permissible limit. The person breaching this rule is entitled for either five year imprisonment or fine up to Rs. 1 lakh. National Green Tribunal has been established in 2010 for effective implementation of environment related laws.

Let's Discuss

Collect the information about Chipko Movement and discuss between two groups of your class about its importance in present situation.

Internet is my friend

1. Sound Pollution (Control & Prevention) Rule, 2000.
2. Biomedical Waste (Management & Handling) Rule, 1998.
3. E-waste (Management & Handling) Rule, 2011.

Always remember

As per wildlife protection Act 1972

As per clause 49 A, trading of rare animals has been completely banned.

As per clause 49 B, use of articles prepared from skin or organs of wild animals has been banned.

As per clause 49 C, disclosure of the stock of artefacts made form rare wild animals is compulsory.

The big story of a small man

Jadav Molai Payeng is a highly capable person in Assam. Born in 1963, he is working as a forest worker since the age of 16 years. Once, large number of snakes died in the flood of Brahmaputra River flowing by the village. As a preventive measure, Molai planted 20 bamboo plantlets. In 1979, the local Social Forestry Department began a social afforestation project on 200 hectares of land. 'Molai' was one of the few forest workers who were looking after that project. Molai continued to plant the trees even after completion of the project. As a result of his continuous work of planting and caring for the trees, the barren area witnessed the forest cover over the 1360 acres.

Today, this jungle in Kokilamukh of Jorhat district of Assam is the result of the hard work for 30 years. He has been awarded with the prestigious 'Padmashree' award by government of India for this unparallel work. Now, it is well known as 'Molai Jungle'. Many people come together to destroy the forest, **but a single person, if determined, can establish a new forest!**

Environmental Conservation and Biodiversity

Most harmful effect of the environmental pollution occurs on the living organisms. Have you seen some examples of this in your area? Our living world had been richly diverse. It consisted of varieties of plants and animals. However, we are not able to see some specific animals about which we had listened from our earlier generations. Who is responsible for this?

Biodiversity is the richness of living organisms in nature due to presence of varieties of organisms, ecosystems and genetic variations within a species. Biodiversity occurs at three different levels.

Genetic Diversity

Occurrence of diversity among the organisms of same species is genetic diversity. Ex. Each human being is different from other. Possibility of wiping out the species arises if there is decrease in the diversity within the species whose members involve in sexual reproduction.

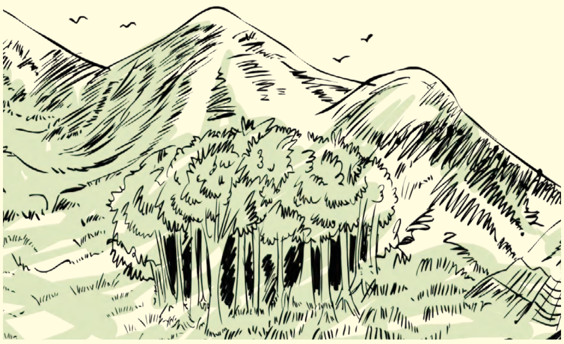
Species Diversity

Innumerable species of organisms occur in the nature. This is called as species diversity. Species diversity includes various types of plants, animals and microbes.

Ecosystem Diversity

Many ecosystems are present in each region. Ecosystem is formed through the interaction between plants, animals, their habitat and changes in the environment. Each ecosystem has its own characteristic animals, plants, microbes and abiotic factors. Ecosystems are also of two types are natural and artificial.

There should be positive attitude of human being towards the environment for welfare of entire living world. For this purpose, following roles are important. You can be a conservator, organizer, guide, plant-friend, etc. Describe about the role you wish to perform and your plans for that role.



4.3 Sacred grove

Sacred Groves

The forest conserved in the name of god and considered to be sacred is called as sacred grove. These are in fact 'sanctuaries' conserved by the society and not by the government forest department. As it has been conserved in the name of god, it has special protection. These clusters of thick forests are present not only in Western Ghats of India but in the entire country.

More than 13000 sacred groves have been reported in India. Where are such sacred groves in Maharashtra? Make a list and visit with your teachers.

Enlist and discuss

Some symbols are given below. Find the meaning of those symbols in relation to environment conservation. Make a list of other such symbols.

Stick here a symbol known to you.

How can biodiversity be conserved?

1. Protecting the rare species of organisms.
2. Establishing national parks and sanctuaries.
3. Declaring some regions as 'bioreserves'.
4. Projects for conservation of special species.
5. Conserving all plants and animals.
6. Observing the rules.
7. Maintaining record of traditional knowledge.

Till now, we have studied the rules and regulations about environmental conservation and protection, in this lesson. Many people in the society are voluntarily coming together to perform this noble work. Many institutes at state, national and international level are involved in this work.

Voluntary Organizations

1. Bombay Natural History Society, Mumbai.
2. CPR environment group, New Chennai.
3. Gandhi Peace Foundation, Environment Cell, Delhi.
4. Chipko Centre, Tehri Garhwal.
5. Centre for Environment Education, Ahmadabad.
6. Kerala Science Literature Council, Trivandrum.
7. Indian Agro Industries Foundation, Pune.
8. Vikram Sarabhai Community Science Centre, Ahmadabad.

International Environment Organizations

1. International Union for Conservation of Nature (IUCN), Gland VD, Switzerland.
2. Intergovernmental Panel on Climate Change (IPCC), Geneva.
3. United Nations Environment Program, Geneva.
4. World Wildlife Fund, New York.
5. Bird Life International, Cambridge.
6. Green Climate Fund, Songdo, S. Korea

Green Peace is world's largest organization engaged in environmental activities. More than 25 lakh people from 26 different countries are members of this organization. Collect more information about the work of above mentioned organizations.

Think and Answer

Attempts at various levels are performed for conserving environment. Role of the person is defined as per these levels. Some roles are given below. Which role would you like to per form? Why ?

My Role in Environment

Conservation
Conserving the available resources

Control
1. Preventing the harm.
2. Stopping the harmful activities.
3. Changing the mindset.

Production
1. Revival of harmed factors of environment.
2. Attempting innovation.

Preservation
1. Preserving whatever has been leftover.
2. Remedies to prevent further loss.
3. Preserving unknown regions.

Prevention
1. Preventing possible harms.
2. Designing new plans
3. Factors harmful to environment

Awareness
1. Education
2. Guidance
3. Awareness
4. Imitation
5. Organization
6. Participation

Hotspots of Biodiversity

34 highly sensitive biodiversity spots are reported all over the world. Such areas had once occupied 15.7% area of the Earth. At present, 86% of the sensitive areas are already destroyed. Presently, only 2.3% area of the Earth has been left over with sensitive spots. It includes 1,50,000 plant species which are 50% of the world count.

As far as India is considered, out of 135 species of animals, 85 species are found in the jungles of eastern region. About 1,500 endemic plant species are found in western ghat. Out of the total plant species in the entire world, 50,000 are endemic. Collect more information about locations of these hotspots present in the world.

Three Endangered Heritage Places of the Country

The Western Ghat spread over the states of Gujarat, Maharashtra, Goa, Karnataka, Tamilnadu and Kerala these six states has been endangered due to mining industry and search for natural gas. Habitats of Asiatic lion and wild bison of this region have been under threat.

Manas National Park of the Assam is under threat due to dams and indiscriminate use of water. Tiger and rhino of that region are under threat.

Sunderban National Park of West Bengal is reserved for tigers. However, the tiger population and overall local environment is seriously challenged by dams, deforestation, excessive fishing, trenches dug for same, etc.

Recall a little Collect the names of extinct birds and animals of India and tell those names to others.

Classification of Threatened Species

1. Endangered Species

Either number of these organisms is declined or their habitat is shrunk to such an extent that they can be extinct in near future if conservative measures are not implemented. Example, Lion tailed monkey, lesser florican.

4.4 Lion-tailed Monkey

2. Rare Species

Number of these organisms is considerably declined. Organisms of these species being endemic may become extinct very fast. Example, Red panda, Musk deer.

3. Vulnerable Species

Number of these organisms is extremely less and continues to decline. Continuous decline in their number is worrisome reason. Example, Tiger, Lion.

4.5 Red Panda

4. Indeterminate Species

These organisms appear to be endangered but due to their some behavioural habits (like shyness) there is no definite and substantial information. Example, Giant squirrel (Shekru).

Specialty of the Day

22nd May: World Biodiversity Day

Survey the plants and animals in your area. Maintain a record about their characteristics.

Do you know?

International Union for Conservation of Nature (IUCN) prepares the 'Red List' that contains the names of endangered species from different countries. Pink pages of this book contain the names of endangered species while green pages contain the names of previously endangered but presently safe species.

Always remember

Always Remember

Let us remember.... Let us behave accordingly......

1. Destroying a plant is to destroy everything.
2. Practice afforestation to conserve environment.
3. Forest is Wealth.
4. Environmental protection is value education.
5. Provident use of paper is prevention of deforestation.
6. To practice the environmental protection is to development of human society.
7. Pure air, pure water is key to healthy life.

Think

World Wildlife Fund (WWF) published a survey in 2008. According to it, about 30% of animal species have become extinct over the period of 35 years (1975 – 2005). What will happen in future if this continues as it is?

Exercise

1. **Reorganize the following food chain. Describe the ecosystem to which it belongs.**
   Grasshopper – Snake – Paddy field – Eagle – Frog.
2. **Explain the statement-'we have got this Earth planet on lease from our future generations and not as an ancestral property from our ancestors.'**
3. **Write short notes.**
   a. Environmental Conservation.
   b. Chipko Movement of Bishnoi.
   c. Biodiversity.
   d. Sacred Groves.
   e. Disaster and its management.
4. **How will you justify that overcoming the pollution is a powerful way of environmental management?**
5. **Which projects will you run in relation to environmental conservation? How?**
6. **Answer the following.**
   a. Write the factors affecting environment.
   b. Why does the human beings have important place in environment?
   c. Write the types and examples of biodiversity.
   d. How the biodiversity can be conserved?
   e. What do we learn from the story of Jadav Molai Peyang?
   f. Write the names of biodiversity hot spots.
   f. Which are the reasons for endangering the many species of plants and animals? How can we save those?
7. **What are the meanings of following symbols? Write your role accordingly?**

Project:

Make a presentation on pollution of Ganga and Yamuna Rivers and effects of air pollution on Tajmahal.

KYD6YE

-: A Pledge for Life :-

I am aware that the diversity on the Earth is for the existence of me, my family and the entire mankind. I am aware about the responsibility of conserving and protecting the rich diversity. I am aware about the fast declining number of wildlife, plants and animals. I am accepting the responsibility of judicious use of natural resources and management of biodiversity.

I pledge for adopting the following principles for happy and satisfactory life of all organisms on the Earth.

I will always try for conservation and sustainable management of natural resources.
I will make the change that I am expecting.
I will be committed for safety of entire life on the Earth.
I will educate the people about benefits of conservation and co-existence.

5. Towards Green Energy

- Use of various energy sources
- Generation of electrical energy
- Process of generation of electricity and environment

Can you recall?

1. What is Energy?
2. What are different types of Energy?
3. What are different forms of Energy?

Let's Discuss

Make a list of the work that we do in our day to day life using energy. Which forms of energy do we use to do this work? Discuss with your friends.

Energy and use of energy

In modern civilization, energy has become a primary need along with food, cloth and shelter. We need energy in different forms for diverse types of works. The energy that we need may be in the form of mechanical energy, chemical energy, sound energy, light energy or heat energy. How do we get these different forms of energy?

Make a table Make a table based on forms of energy and corresponding devices.

We know that energy can be converted from one form to another. Different sources of energy are used to the different forms of energy necessary for us. In previous standards we have learnt about energy, sources of energy and various concepts related to them. Here we will learn about various sources that are now used for the generation of electrical energy, the methods that are used for this, the scientific principles that are used there, the advantages and disadvantages of these methods and also what is meant by green energy.

Can you tell?

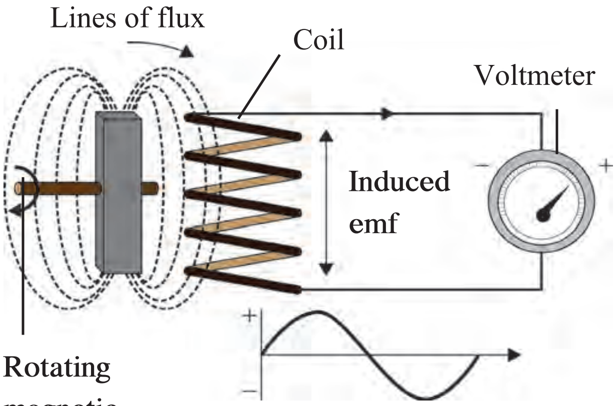
1. Where do we use electrical energy in our day-to-day life?
2. How Electric energy is produced ?

Generation of electrical energy

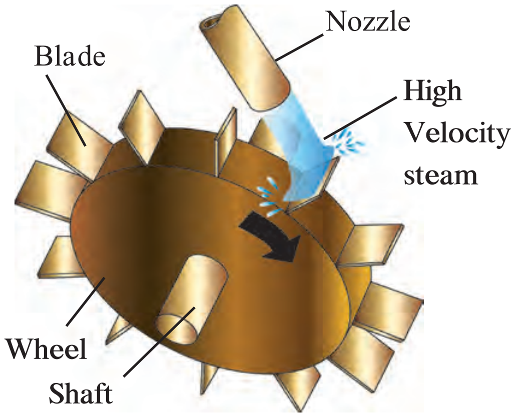
Most of the electric power plants are based on the principle of electromagnetic induction invented by Michael Faraday. According to this principle, whenever magnetic field around a conductor changes, a potential difference is generated across the conductor.

The field around a conductor can be changed in two ways. If a conductor is stationary and magnet is rotating, the field around the conductor changes or if a magnet is stationary, but the conductor is moving then also the field around the conductor will change. Thus, in both these cases, a potential difference is created across the conductor. (Figure 5.1). The electrical power generating machine based on this principle is called electric generator.

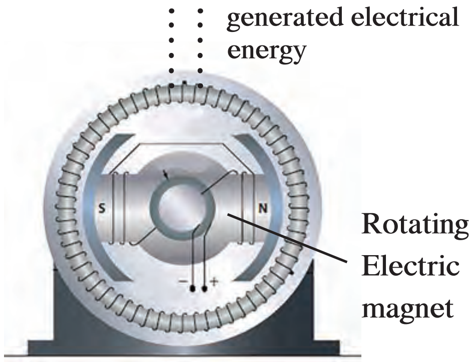
Such large generators are used in commercial power generation plants. Turbine is used to rotate the magnet in the generator. A turbine has blades .When a flow of liquid or gas is directed on the blades of the turbine, it rotates (see Figure 5.2). because of the kinetic energy of the flow.This turbine is connected to electric generator. Thus the magnet in electric generator starts rotating and electric energy is produced (Fig.5.3)



5.1Electromagnetic Induction



5.2 Steam turbine



5.3 Schematic of electric generator

This method of electric energy generation can be represented as below.

Thus, to generate electricity based on the principle of electromagnetic induction, we need a generator. To rotate the generator we need a turbine and to drive the turbine, we need an energy source. Based on which type of energy source is used to rotate the turbine, there are different types of power generating stations. The design of the turbine used in different types of power stations is also different.

Proper energy source to run turbine → **Turbine** → Generator → Electrical energy

5.4 Flow chart showing generation of electrical energy

Thermal energy based electric power station

In this the turbine is rotated using steam. Water is heated in a boiler. Using the thermal energy released due to burning of coal. Steam of very high temperature and pressure is generated. The energy in the steam drives the turbine. Thus, the generator connected to the turbine rotates and electrical energy is produced. The steam is converted back into water and the water is re-circulated to the boiler. This is shown in flow chart in fig 5.5

Fuel Coal → Boiler for conversion of water to steam → Steam turbine → Generator → Electrical energy

Steam turbine ↓ System for converting steam back to water ↑ Boiler for conversion of water to steam

5.5 Flow chart showing generation of electrical energy using thermal energy

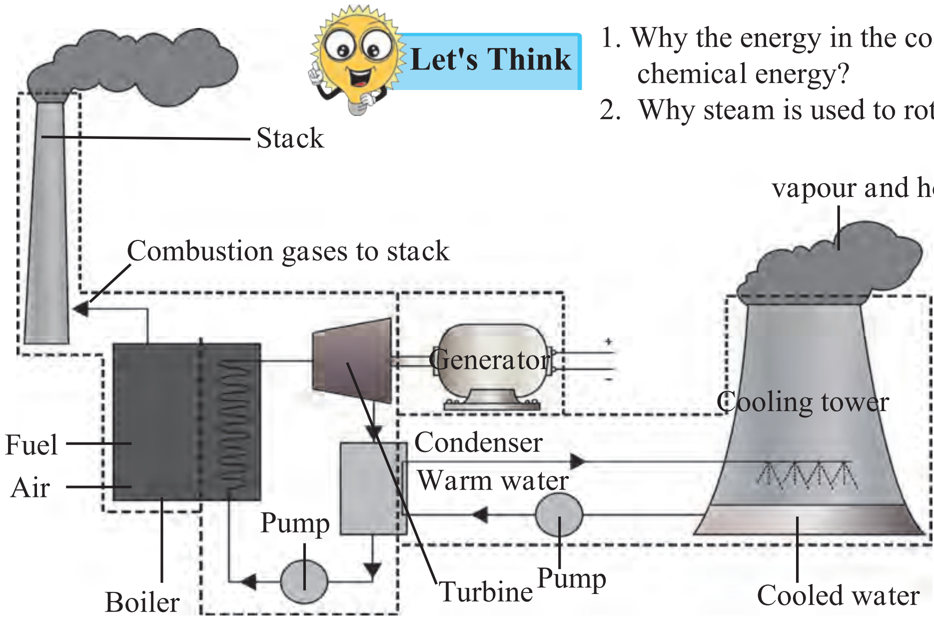
Since thermal energy is used here to generate electrical energy, such power plants are called thermal power plants. In thermal power plants, the chemical energy in the coal is converted into electrical energy through several steps which are shown in figure 5.6.

Chemical energy in coal → Thermal energy → Kinetic energy in steam → Kinetic Energy in turbine → Electrical energy

5.6 Energy transformation in thermal power plant

Let's Think

1. Why the energy in the coal is called as chemical energy?
2. Why steam is used to rotate the turbine?



5.7 Schematics of Thermal power plant

If you see a thermal power station, you will observe two types of towers there. What are they? If you observe the schematic of the thermal power station in Figure 5.7 , you will get answer to this question.

Compare the schematic of the thermal power station with the block diagram above and you will understand how the boiler, turbine, generator and the condenser are arranged in the power station.

After combustion of fuel (here, coal) in the boiler, the emitted gases are released to the atmosphere through very high tower. Once the turbine is rotated using the steam at high temperature and high pressure, steam temperature and pressure decreases. This steam is converted back to water by taking out heat from it (i.e.by cooling it) . This is done in the condenser using water in the cooling tower. The water in cooling tower is circulated through the condenser. Heat energy in the steam is given to the water and the steam condenses back to water. The heat absorbed by the water is then released to atmosphere through vapour and heated air through cooling tower. Although, thermal power generation is a major way of electricity generation today, it suffers from certain problems

Use of ICT

Prepare a presentation about thermal power plant using computerized presentation, animation, video, pictures, etc. Send it to others and upload on You Tube.

Problems

1. Air pollution due to burning of coal: Burning of coal results in emission of gases like carbon dioxide, sulphur oxide and nitrogen oxide which are harmful to the health.
2. Along with the emission of gases due to burning of coal, soot particles are also released into the environment. This may cause serious health problems related to the respiratory system.

Do you know?

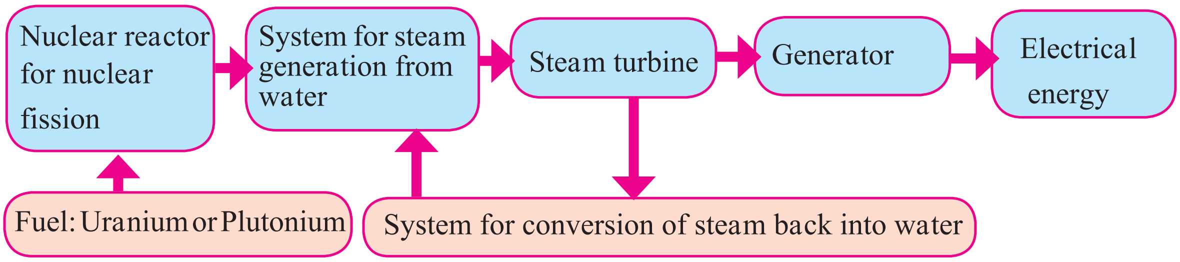
Some major thermal power plants in India and their capacity

Place	State	Capacity (MW)
Vindhyanagar	Madhya Pradesh	4760
Mundra	Gujarat	4,620
Mundra	Gujarat	4,000
Tamnaar	Chhattisgarh	3,400
Chandrapur	Maharashtra	3,340

3. The reserves of fuel used in this method i.e. coal are limited. Therefore, in future, there will be limitations on the availability of the coal.

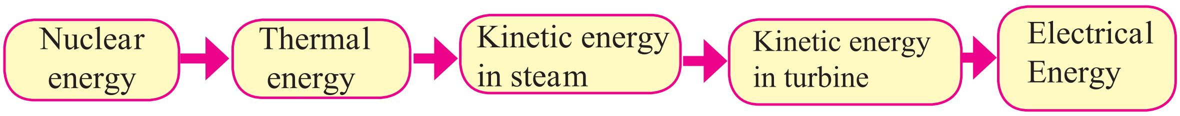
Power plant based on Nuclear Energy

In the power plant based on nuclear energy also , steam turbine is used to rotate the generator. However, here, the energy released by fission of nuclei of atoms like Uranium or Plutonium is used to generate the steam of high temperature and high pressure. The energy in the steam rotates the turbine, which in turn drives the generator producing electricity. The flow chart of nuclear power plant is shown in fig 5.8 .



5.8 Nuclear power plant

Thus, here nuclear energy is converted into thermal energy, thermal energy is converted into kinetic energy of steam, kinetic energy of steam is converted into kinetic energy of turbine and finally the kinetic energy of the turbine is converted into electrical energy .The step-by-step transformation of energy is shown in figure 5.5.



5.9 Energy transformation in nuclear power plant

Can you tell? How does nuclear fission take place?

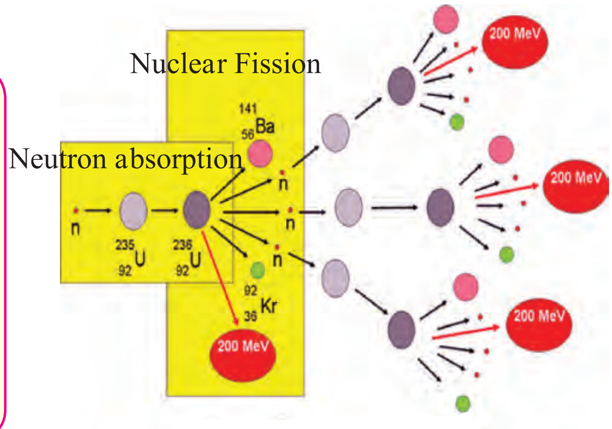
When neutron is bombarded on atom of Uranium - 235 , it absorbs the neutron and converts into its isotope Uranium - 236. Uranium - 236 being extremely unstable converts into atoms of Barium and Krypton through a process of fission releasing three neutrons and 200 MeV energy. The three neutrons generated in this process cause fission of three other Uranium - 235 atoms releasing more energy.

The neutrons released in this reaction release more energy through fission of more uranium nuclei. This process of fission of Uranium -235 atoms continues and is called the chain reaction. In nuclear power plants, a controlled chain reaction results in release of thermal energy, which is used for electric energy generation.

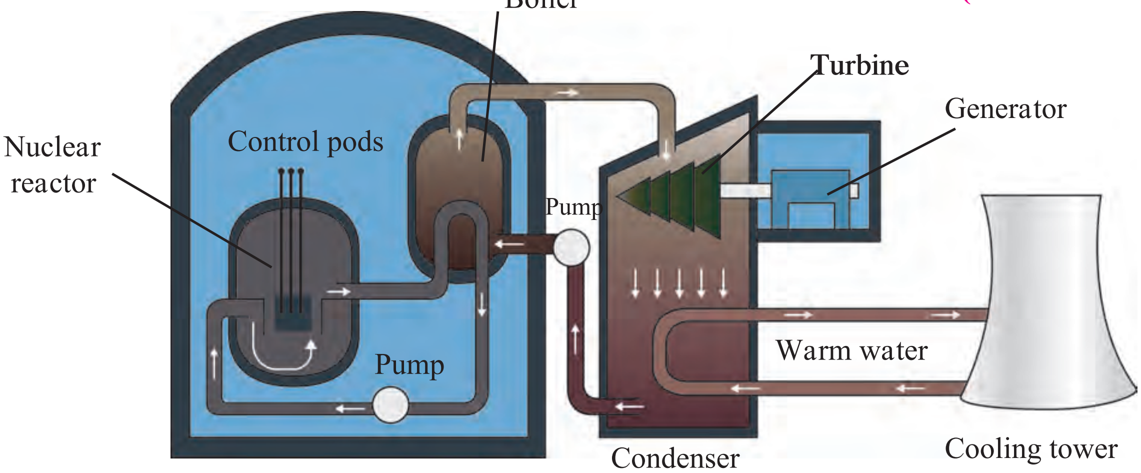
Internet is my friend

Complete the following table for some important nuclear power plants in India.

Place	State	Capacity (MW)
Kudankulam	....	....
Tarapur	....	....
Ravatabhata	....	....
Kaiga	....	....



5.10 Nuclear fission (Chain reaction)



5.11 Schematic diagram of nuclear power plant

A nuclear power plant does not use fossil fuel like coal. Therefore, problems like air pollution do not arise. Also, if sufficient nuclear fuel is available, this can be a good source of electrical energy. However, there are few problems associated with nuclear power generation.

Problems:

1. The products after fission of nuclear fuel are also radioactive and emit harmful radiations. The products are called as nuclear waste. How to dispose the nuclear waste safely is a big challenge before the scientists.
2. An accident in nuclear power plant can be very fatal. This is because the accident may result in release of very harmful radiations.

Compare

Observe the schematic of thermal power plant and the nuclear power plant. Discuss what are the similarities and differences between the two?

Power generation plant based on energy of natural gas

In this plant, the turbine is run by a gas at very high temperature and pressure generated by combustion of natural gas. A flow chart showing various stages in the power generation plant based on natural gas energy is shown in figure 5.12.

Compressor for pressurised air → Combustion chamber → Gas turbine → Generator → Electrical energy

Natural gas → Combustion chamber

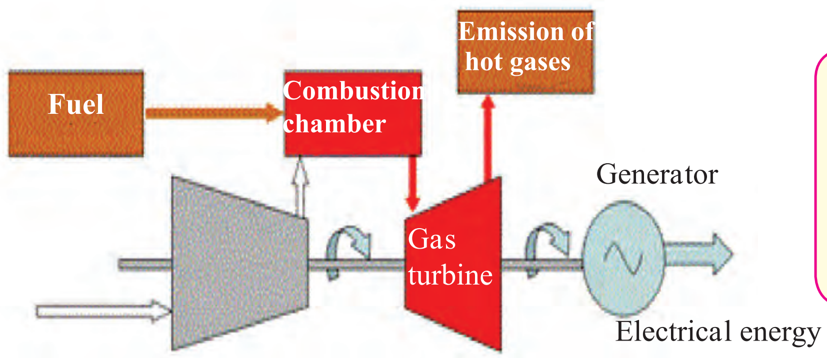
5.12 Arrangement in power plant based on energy of natural gas

There are three main sections in this type of plant. Pressurised air is introduced into the combustion chamber using a compressor. In the combustion chamber the natural gas burns in presence of the air. The gas at very high temperature and pressure generated in this chamber runs the turbine. The turbine then drives the generator to produce electricity. Step-by-step transformation of energy in this plant is shown in fig 5.13.

Chemical energy in natural gas → Kinetic energy generated due to combustion of natural gas → Kinetic energy in turbine → Electrical energy

5.13 Transformation of energy in power plant using energy of natural gas

The efficiency of this type of power generation plant is higher than that of power generation plant based on coal. Moreover, since the natural gas does not contain sulphur, burning of natural gas results in less pollution. The schematic of power plant based on natural gas is given in figure 5.14.



5.14 Schematic of power plant based on natural gas

Let's Think

Which electricity generation process is eco-friendly and which not?

Some natural gas based power plants and their capacity

Place	State	Capacity(MW)
Samaralkota	Andhra Pradesh	2620
Anjanvel	Maharashtra	2,220
Bavanaa	Delhi	1,500
Kondapalli	Andhra Pradesh	1466

Always remember

Though use of energy is unavoidable in our day to day life, it is necessary to use it carefully and only in the required amount.

Electric energy generation and environment

Electricity generation based on fossil fuels like coal, natural gas and nuclear fuels like uranium and plutonium are not environment friendly. It means, that if electrical energy is generated using these fuels, it can lead to environmental degradation.

1. We have seen that burning of fossil fuels like coal, and natural gas leads to emission of certain gases and soot particles. This results in air pollution. Incomplete combustion of fuels leads to formation of carbon monoxide. It adversely affects our health. Increase in percentage of carbon dioxide in the air due to burning of fuels affects environment severely. The phenomena of global warming is an example of this. Nitrogen dioxide generated due to burning of fuels like coal, diesel, petrol, etc. leads to problems like acid-rain. Soot particles generated due to incomplete burning of fossil fuel cause air pollution. It can lead to problems related to respiratory system, like asthama.
2. It took millions of years for formation of fossil fuels like coal, crude oils and natural gases (LPG and CNG). Also, the reserves of these fuels are limited. They are going to deplete in future. It is said that with the current speed of their use, the coal reserves in the world would last for another about 200 years or so and the natural gas reserves for about 200-300 years.
3. We have also discussed above about the problems in use of nuclear energy like the disposal of nuclear waste and possibility of disaster due to accident in nuclear power plant.

Considering all these points, it can be said that the energy generation from fossil fuels and nuclear fuels are not environment friendly.

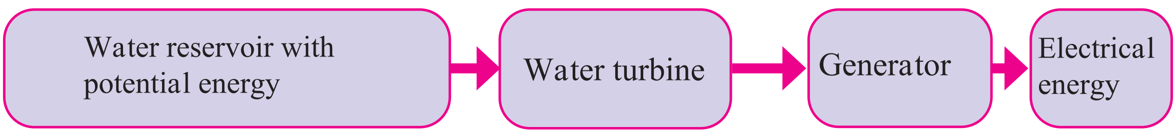
Towards environment friendly energy…..towards green energy:

There are other ways of electricity production which avoids above problems. Electricity generation from water reservoir, wind, sunlight, biofuels etc. are the examples of such methods. The energy sources used in such options i.e. water-reservoir, wind, sunlight, biofuel are never-ending i.e are perpetual. Moreover, use of these sources do not lead to environmental problems discussed above. Therefore, electricity generation through these sources can be called environment friendly. We can also call the energy generated by these processes as green energy. Looking at the problems in electricity generation using fuels like coal, natural gas and nuclear fuels, the world is now heading towards environment friendly energy i.e. green energy.

Hydroelectric Energy

Kinetic energy in flowing water or the potential energy in water reservoir is a conventional source of energy. In hydroelectric power plant, the potential energy in water stored in dam is converted into kinetic energy of water. Fast flowing water is brought from the dam to the turbine at the bottom of the dam. The kinetic energy of the flowing water drives the turbine. The turbine in turn drives the generator to generate electricity.

The block diagram showing different components of hydroelectric power plant is shown in figure 5.15



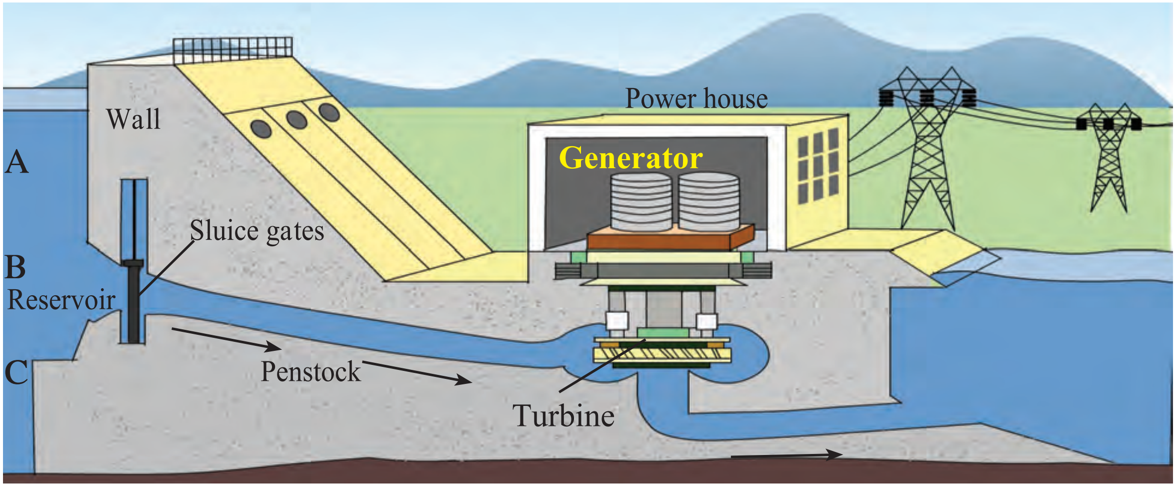
5.15 Different stages in hydroelectric power plant

Electricity generation using wind-energy

Potential energy in water → Kinetic energy in flowing water → Kinetic energy in turbine → Electrical energy

5.16 Energy Transformation in hydroelectric power plant

The schematic of hydroelectric plant is shown in Figure 5.17. Water from about middle of the total height of the dam is taken to the turbine, as shown by point B in the diagram.



5.17 Schematic diagram of Hydroelectric power plant

Use your brain power

1. With reference to point B, potential energy of how much water reservoir in the dam will be converted into kinetic energy?
2. What will be the effect on electricity generation, if the channel taking water to turbine starts at point A?
3. What will be the effect on electricity generation, if the channel taking water to turbine starts at point C?

Since no fuel is burnt in hydroelectric plant, no air pollution due to combustion of fuel results. However, considering the issues like forced migration of large community, submerging of forests and fertile land, adverse effect on living creatures in the river, it has always been a point of debate whether the hydroelectricity is environment friendly or not. What is your opinion about it?

Advantages of hydroelectric power generation

1. Since no fuel is burnt in hydroelectric power generation, there is no pollution resulting from combustion of fuels.
2. If there is sufficient water storage in the dam, it is possible to generate electricity as and when necessary.
3. Although water reservoir is used for power generation, it can be replenished during rainy season leading to uninterrupted power generation.

Problems associated with hydroelectric power plant

1. The back-water due to storage of water in dam may submerge villages or towns in that area. This leads to the problems of rehabilitation of the displaced population. Moreover, this can also submerge forests as well as fertile land.
2. The obstruction of the flow of river water may have adverse effect on living world in the river.

Do you know?

Some major hydroelectric plants in India and their capacity (MW)

Place	State	Capacity (MW)
Tehari	Uttarakhand	2400
Koyana	Maharashtra	1960
Srishailam	Andhra Pradesh	1670
Nathpa Zakri	Himachal Pradesh	1500

5.18 Koyana Dam

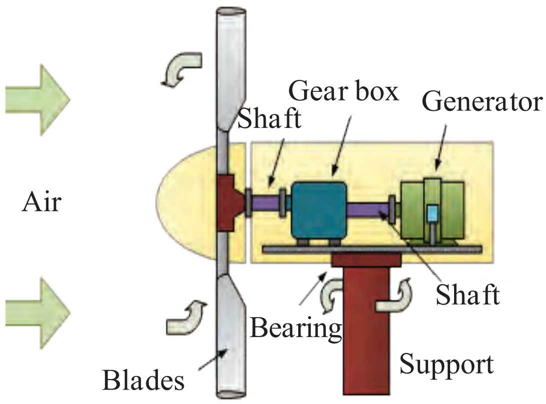
Find out

What is lake tapping? Why it takes place?

The kinetic energy in wind has been used since long for lifting of water, for driving floor mill etc. The wind energy can also be used for electricity generation. The machine which converts the kinetic energy of wind to electrical energy is called wind-turbine. As the wind strikes the blades of the turbine, the blades rotate. The axel of the turbine is connected to electric generator through a gear-box. The function of the gear-box is to increase the rotations per unit time. Thus, the rotating blades drive the turbine and the turbine in turn drives the generator to generate electricity. Various stages in the wind-energy generation system can be shown in figure 5.19 and schematics of a wind mill is shown in figure 5.20.

Wind with required velocity → Wind Turbine with large fins → **Generator** → Electrical energy

5.19 Stages in electric generator using wind energy



5.20 Schematic of wind mill

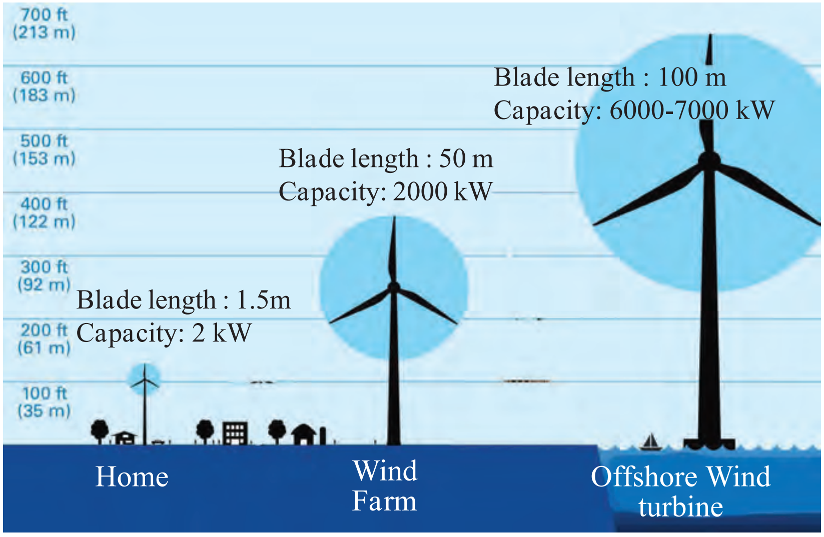
The energy conversion process is shown in figure 5.21.

Kinetic Energy in wind → **Kinetic energy in turbine** → **Electric energy**

5.21Transformation of energy in an electric generator using wind energy

Wind turbines with capacity right from less than 1 kW to about 7 MW (7000 kW) are commercially available. Depending on the wind velocity available at the site of installation, wind-turbine with specific capacity is selected. The wind velocity at specific location depends on many geographical factors.

Wind velocity is usually high on sea shores and that environment is appropriate for installation of wind turbine. Wind-energy is a clean energy source. However, the wind-velocity necessary for wind-energy generation is not available everywhere. In that sense, use of wind-energy is limited.



5.22 Wind turbines of different capacities

Get information

Get information about major wind-power stations in India and their capacity. Make a table of their location, state and their power generation capacity in MW.

Electric Energy generation using solar energy

Using the energy in the Sunlight, electric energy can be generated in two ways:

1. In all the above methods of electricity generation we have studied, the electric generator is driven by using some source of energy and electricity is generated by making use of the principle of electromagnetic induction. However, electrical energy can be generated directly from solar radiation without using generator and without using the principle of electromagnetic induction. This happens in solar photovoltaic cells. Solar photovoltaic cells convert the solar energy directly into electrical energy.

2. In the second method, the energy in solar radiation is converted into thermal energy first. Then a turbine-generator system is driven using that thermal energy to generate electricity.

1.Solar photovoltaic cell

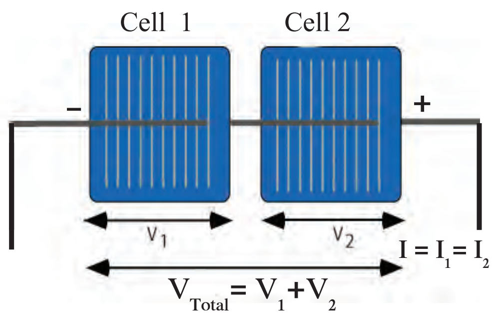
Solar photovoltaic cell converts the solar radiation energy directly into electrical energy. This is called solar photovoltaic effect. The electrical energy generated through this energy transformation process is DC in nature. These solar cells are made of a special type of material called semiconductor (e.g. silicon). A silicon solar cell of dimension 1 cm^2 generates current of about 30 mA and potential difference of about 0.5 V. Thus, a silicon solar cell of dimension 100 cm^2 will generate about 3 A (30 mA/cm^2 X 100 cm^2=3000 mA= 3 A) current and 0.5 V. Remember that the potential difference available from a solar cell is independent of its area.

Energy in Sunlight $\xrightarrow{\text{Photovoltaic effect}}$ **DC Electric power**

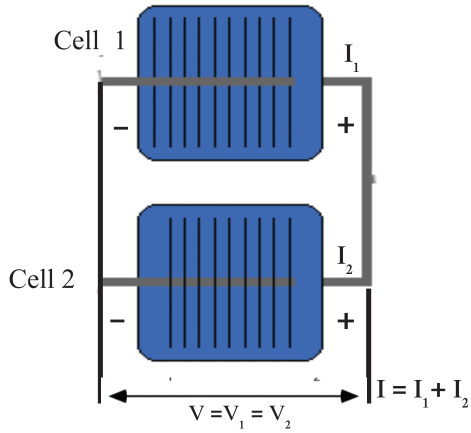
If two solar cells are connected in series as shown in figure 5.23, the potential difference obtained from this combination is addition of the potential differences of individual solar cells. However, the current generated from this combination is equal to the current from an individual cell. It means that when solar cells are connected in series, currents from the individual cells are not added. Similarly as shown in figure 5.24, if two solar cells are connected in parallel, the current generated from this combination is the summation of the currents from an individual solar cells. However, the potential difference obtained from this combination is the same as the potential difference obtained from individual cell. Thus, if two solar cells are connected in parallel, the potential differences from the two cells are not added.

In this way, by connecting many solar cells in series and in parallel solar panels generating required current and potential difference are made. See Figure 5.25. For example, if 36 solar cells, each of size 100 cm^2 are connected in series in a solar panel, it will give potential difference of 18 V and current of 3 A. Many such panels are connected together to generate electricity on larger scale. A good solar cell can have an efficiency of around 15%. It means that if a solar panel receives power of 100 watt from solar radiation, the electrical power output from the panel will be 15 watt.

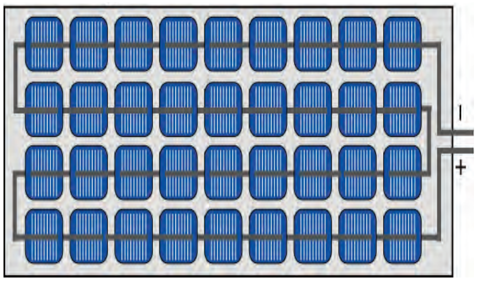
Many solar panels are connected in series and in parallel to generate required current and potential difference. As shown in Figure 5.26, solar cell is the basic unit in solar electric plant . Many solar cells come together to form a solar panel. Many solar panels connected in series form a solar strings, and, many solar strings connected in parallel form a solar array. As we can obtain as much electrical power as needed, they are used in applications which need marginal power (e.g. calculators that run on solar energy) to power station of MW capacity.



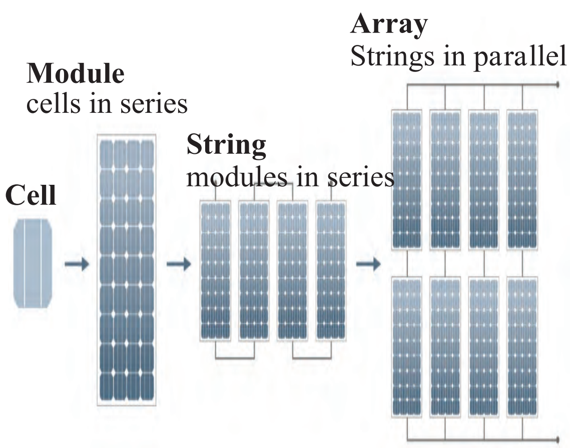
5. 23 Solar cells in series



5.24 Solar cells in parallel



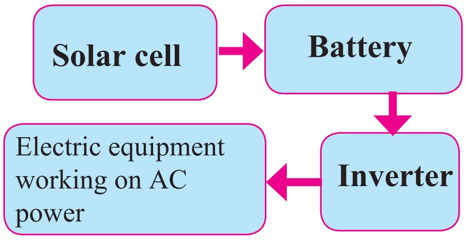
5.25 A solar panel made from 36 solar cells



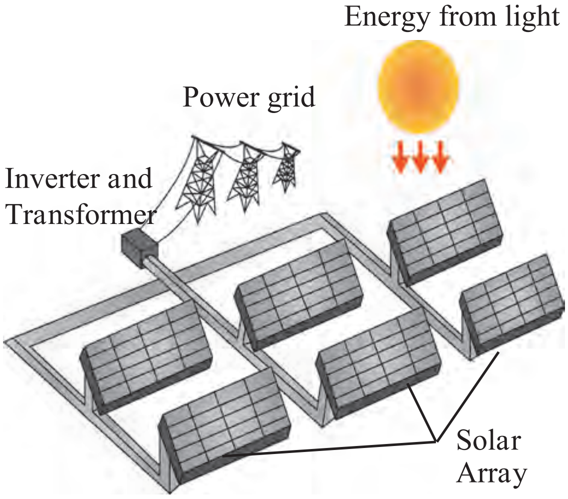
5.26 Solar cell to solar array

The power available from the solar cells is DC. So, in applications which need DC power , e.g. electric lights based on Light Emitting Diodes, the energy can be directly used. However, since the energy from solar cell is available only in presence of sunlight, the energy has to be stored in batteries for use at later time.

However, most of the equipment in domestic as well as industrial use run on AC power. In such case, the DC solar power must be converted to AC power using an electronic device called inverter(Figure 5.27).



5.27 Conversion of energy generated by cells to AC form by using inverter



5.28 Schematic of solar photovoltaic station

We have seen that many solar panels can be connected together to generate whatever energy we need. As shown in Figure 5.28, the DC power generated from these panels is first converted into AC power. A transformer transforms the voltage and current levels of the generated power and then it is fed into the electricity distribution network. Figure 5.28 is a schematic diagram of solar photovoltaic power station.

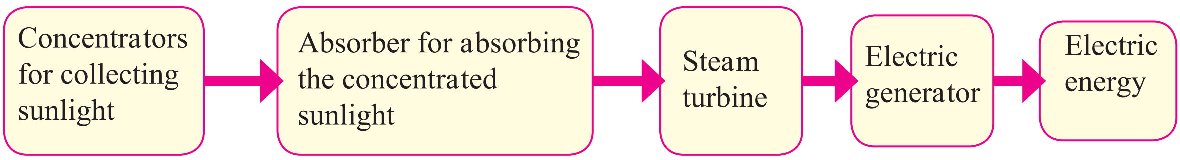
Find out

Gather information about major solar photovoltaic power generating plants and their capacity in India.

In this way, electricity is generated without any fuel combustion and so without any air pollution. However, since the energy is generated using solar radiation, solar cells can generate electricity during day-time only.

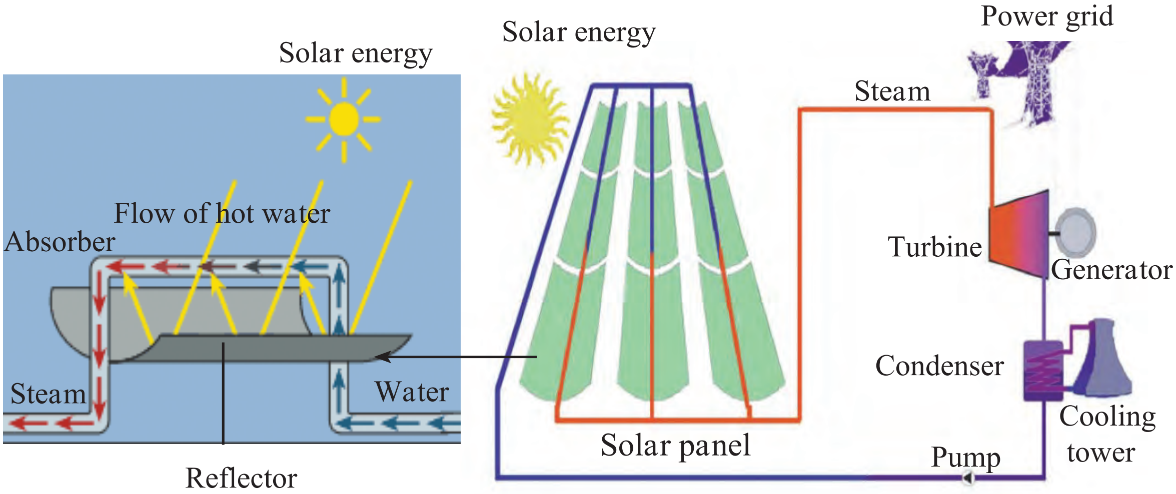
Solar Thermal power plant

We have seen that thermal energy generated from coal and nuclear fuel can be used to generate electricity. Thermal energy can also be generated from solar radiation and can be used for electricity production. Different stages in such solar thermal power plant are as shown in figure 5.29



5.29 Different stages in solar thermal power plant

As shown in Figure 5.30 , many reflectors reflect and concentrate solar radiation on absorbers. There solar energy is converted into heat energy. Using this heat energy steam is generated to drive the turbine and generator.



5.29 Schematic of solar thermal power plant

Do you know?

Energy sources use for electrical power generation in the world.

Sources	**World (%)**	**India (%)**
Coal	41	60
Natural Gas	22	08
Hydroelectric	16	14
Nuclear energy	11	02
Petroleum	04	0.3
Renewable sources (wind, solar etc.)	06	15.7
Total	**100**	**100**

Exercise

1. **Remake the table taking into account relation between entries in three columns.**

I	II	III
Coal	Potential energy	Wind electricity plant
Uranium	Kinetic Energy	Hydro electric plant
Water Reservoir	Nuclear Energy	Thermal plant
Wind	Thermal Energy	Nuclear power plant

2. **Which fuel is used in thermal power plant? What are the problems associated with this type of power generation?**
3. **Other than thermal power plant, which power plants use thermal energy for power generation? In what different ways is the thermal energy obtained?**
4. **Which type/types of power generation involve maximum number of steps of energy conversion? In which power generation is the number minimum?**

5. Solve the following crossword puzzle.

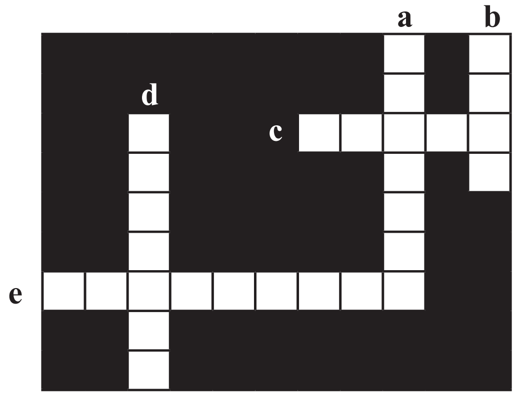
a. Maximum energy generation in India is done using..... energy.

b. ...... energy is a renewable source of energy

c.Solar energy can be called.... energy.

d .... energy of wind is used in wind mills.

e. ..... energy of water in dams is used for generation of electricity.



6. Explain the difference.

a. Conventional and Non-conventional Sources of energy.

b Thermal electricity generation and solar thermal electricity generation.

7. What is meant by green energy? Which energy sources can be called as green energy sources and why? Give example.

8. Explain the following sentences.

a. Energy obtained from fossil fuels is not green energy.

b. Saving energy is the need of the hour.

9. Answer the following questions.

a. How can we get the required amount of energy by connecting solar panels?

b. What are the advantages and limitations of solar energy?

10. Explain with diagram step-by-step energy conversion in

a. Thermal power plant

b. Nuclear Power Plant

c. Solar thermal power plant

d. Hydroelectric power plant

11. Give scientific reasons

a. The construction of turbine is different for different types of power plants.

b. It is absolutely necessary to control the fission reaction in nuclear power plants.

c. Hydroelectric energy, solar energy and wind energy are called renewable energies.

d. It is possible to produce energy from mW to MW using solar photovoltaic cells.

12. Draw a schematic diagram of solar thermal electric energy generation.

13. Give your opinion about whether hydroelectric plants are environment friendly or not?

14. Draw neat and labelled diagrams.

a. Energy transformation in solar thermal electric energy generation.

b. One solar panel produces a potential difference of 18 V and current of 3A. Describe how you can obtain a potential difference of 72 Volts and current of 9 A with a solar array using solar panels. You can use sign of a battery for a solar panel.

15. Write short note on

Electrical energy generation and environment.

Project :

1.Gather information about solar light, solar water heating system and solar cooker.

2.Gather information about a power plant near your locality by vising the plant.